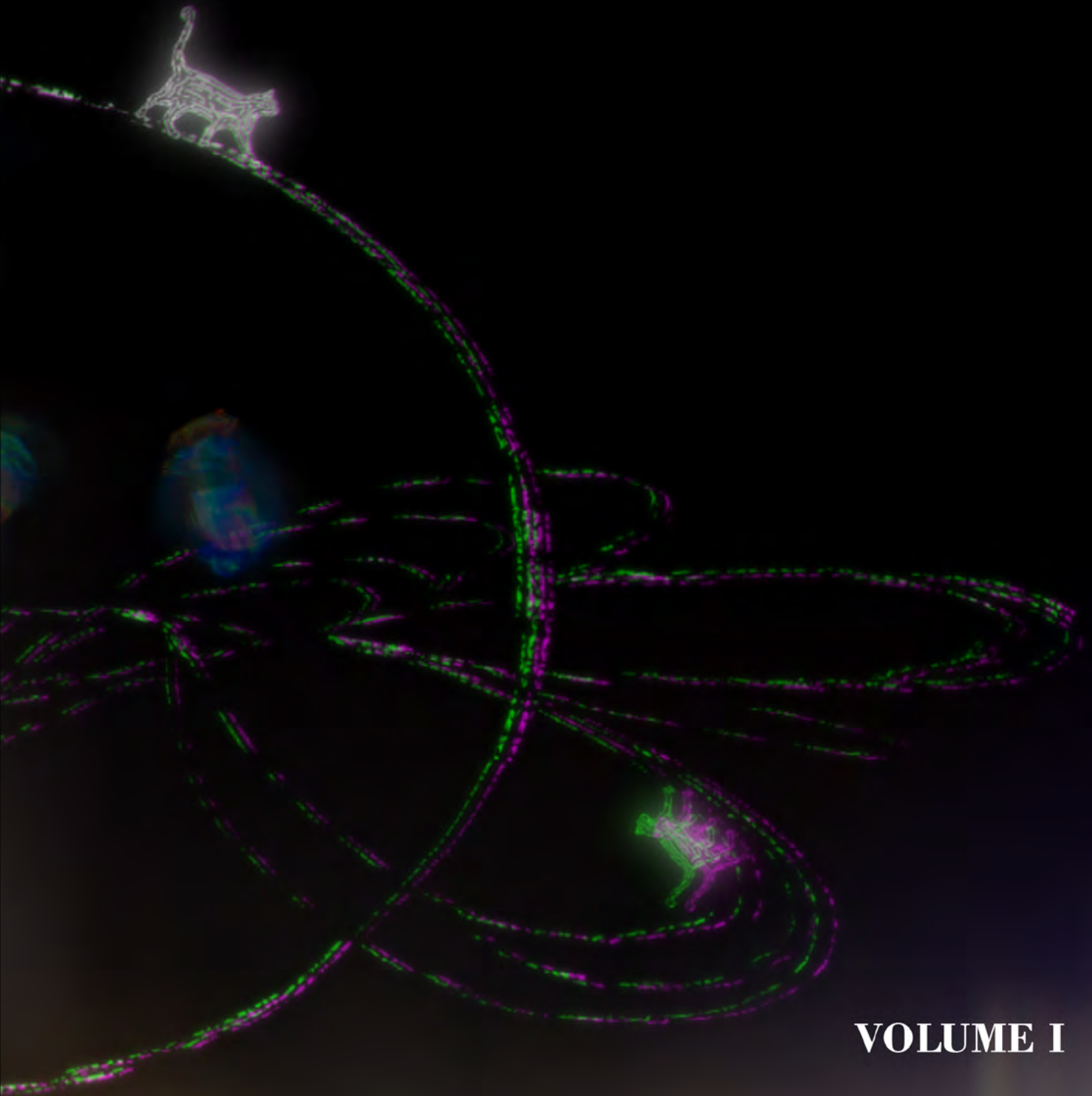


entangled

[ɪn'tæŋɡəld] magazine

some call it curiosity, some call it chaos
psi(t) calls it *entangled*.



VOLUME I

The Universe is, and we are.

Solanum, an observer

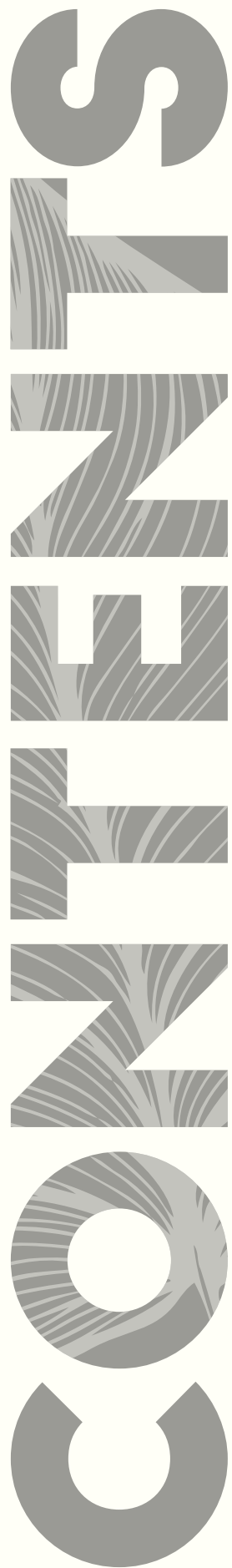
INAUGURAL EDITION

ENTANGLED

$\Psi(t)$

Physics Society of
IISER THIRUVANANTHAPURAM





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03	PATHFINDER An introduction to biophysics - where physics and mathematics help unravel the complexity of living systems. From embryonic pattern formation to cardiac dynamics, this Pathfinder offers students a glimpse into a field where equations meet life itself.
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38	STARS AND BEYOND A compelling journey through stellar nurseries and newborn planetary systems, blending astrophysics and personal research to explore how stars, dust, and cosmic voids shape the very foundations of life.
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47	THE EVENT HORIZON As Dr. Nitin Yadav bids farewell to IISER Thiruvananthapuram, we look back at a journey through solar plasma, magnetic fields, and the dynamics of our nearest star. From coronal waves to computational astrophysics, this conversation offers a glimpse into the physics - and people - behind heliophysics research.
49	EVOLVING PATTERNS As Dr. Chandrakala Meena moves on from IISER Thiruvananthapuram, we revisit a journey through nonlinear dynamics, complex networks, machine learning, and biological systems. From chaos and pattern formation to collaboration and curiosity, this conversation reflects on the interdisciplinary spirit behind her research and mentorship.
52	SEEING BENEATH THE SKIN Blending optics and acoustics, photoacoustic imaging transforms pulses of light into sound to reveal structures hidden deep within biological tissue. Combining the contrast of optical imaging with the depth of ultrasound, it opens new possibilities for non-invasive medical diagnostics and functional imaging.
57	LED THERE BE VOLTAGE! A cardboard box, a few LEDs, some PVC pipes, and a multimeter-that was enough to build a working sun photometer. What began as a wildly improvised idea slowly turned into a real experiment, complete with Langley plots, rooftop observations, and the thrill of watching physics emerge from a handmade setup.
59	FIRST LIGHTS From the Cosmic Microwave Background to the first stars and galaxies, the universe's earliest light still carries the story of cosmic beginnings across billions of years. A journey through the dark ages, primordial stars, and the origins of the structures that shaped the cosmos we see today.
62	BLACK HOLE A journey through the strange and captivating world of black holes - from Einstein's equations and stellar collapse to quasars, gravitational waves, and the supermassive giants shaping galaxies across the universe.

THE GUIDING FRAME

DR. SOUMEN BASAK
Faculty Incharge, PSI(t)



DR. SURAJ S HEGDE
Faculty Incharge, PSI(t)



It is a privilege to assume the role of Faculty-in-Charge of the Physics Society, PSI(T), alongside my colleague Dr. Soumen Basak. Since joining the institute, I have had the opportunity to observe several of the society's events and have been impressed by the dedication of its organizing members and the high standard of its activities and invited speakers.

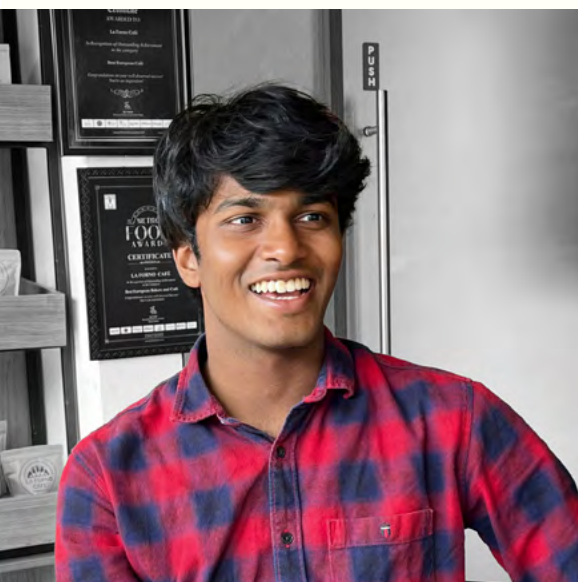
Moving forward, our objective is to sustain this momentum while focusing on initiatives that are valuable not only to students but to the department as a whole. As a student-run organization, PSI(T) is well positioned to broaden students' perspectives on what it means to pursue a degree in Physics

at an institute of national importance, and how this training connects to future academic and societal contributions.

A key priority is to organise events that highlight areas of physics that are underrepresented in the curriculum and less visible on popular platforms. While students are often drawn to widely discussed topics such as black holes or quantum computing, it is essential to expose them to the wider landscape of fundamental research. Such exposure is important for informed choices regarding thesis work and research careers, as well as for understanding the theoretical, computational, and experimental skills required across different fields.

This edition of "entangled" already has a significant wavefunction-overlap on these themes as seen in sections such as "Pathfinder", "Evolving Patterns". In the coming semesters, with the continued efforts, the PSI(T) team hopes to develop programs aligned with these goals. We look forward to active participation from all students in fostering a deeper engagement with the discipline and the broader scientific community.

- Dr. Suraj S Hegde



PRESIDENT'S NOTE

KRTIN PRABHU P
BSMS Batch '22

PSI(t) has played a crucial role in shaping my college life. In my three years at the club, since joining as a volunteer in 2022 to completing my tenure as President in 2025, it has been a very unique and enriching experience. The club has transitioned through more phases than an array of rydberg atoms in this period. Yet, we have managed to find steady states throughout the years. Amidst all of the chaos, the club emerged with the motto to make physics accessible to everyone. We try to inspire students, to take up physics by giving them a chance to explore the subject in all its glory. We show them a simplistic picture of what's beneath the surface, beyond popular science, and help them realise how beautiful core physics truly is.

For a researcher, the most helpful tool in their research is collaboration and exchange of information. At the helm of revolutionary ideas lie interesting discussions between people who share mutual interests. The club strives to bring people with common interests together and give them a platform to discuss such ideas with a larger crowd, where they could potentially expand their cohort. This magazine is yet another venture from the club that aligns with the same goal. We bring you articles from enthusiastic students who have volunteered to write about physics topics that resonate with them, with a hope that it might ignite an interest in one of you readers too. This magazine also embodies the clubs' spirit in simplifying physics for all, guiding young minds through various interesting sections within. I will let you explore it all for yourself without any more spoilers here.

What I have had with PSI(t) cannot be explained in words, it can only be experienced. I have witnessed the club at its peak, at its toughest times, and also its revival. My team and I have poured our efforts into the club for one purpose, to give people a place that they feel they belong to, where they can find a sense of community among people with shared interests. A college environment can be intimidating,

but it gets less daunting when you realise there is a place for you that will help you grow unconditionally. I have come to find people here that have offered wonderful perspectives, not only in physics, but about academics and everything beyond. At the end of the day it's about connecting with people and making the most of these years at IISER TVM, and what better way to go about that than through the shared love for physics.

I have a lot of people to thank, personally for showing me why this club is important and for helping us through various hurdles professionally. Firstly I would like to thank Dr. Vinayak Kamble, our former faculty in-charge. He has been a pillar for us for years and has taught us how to navigate official procedures and operate efficiently. We owe him a great deal. I would like to thank the members of the school of physics and the HOD Prof. Shaijumon for constantly supporting the club and involving us in their activities. I convey my gratitude to Director Prof. JN Moorthy, it's because of his support we have this freedom to conduct so many talks and events. The founders and the former members of the PSI(t) who laid the groundwork and inspired us to be a part of this club deserve all the gratitude we can give. They have been the best people that we could look up to at all times.

Finally I must thank all my friends and fellow committee members for being part of this amazing journey with me. I wish the club the very best, to flourish in the years to come, finding newer, better ways to help students connect with physics and grow together as a community.



FROM THE SECRETARY'S DESK

PATATRA CHOWDHURY
BSMS Batch '22

Hi, this is Patatra. On behalf of the team, I thank you for taking the time and attention to explore our first edition of Entangled.

Being the Secretary of PSIT has been a very eventful and gratifying experience. I started my journey with PSIT by attending the orientation in my first year, then joining as a volunteer and from there now I have hosted two orientations. When I attended these I was sparked to explore and share my interests towards Physics. From there, having had the opportunity to foster and sustain the spark for investigating the world around us, even in a single person is a complete circle. This club has gone through various phase transitions and each one has brought us lessons to learn. What I cherished the most were the interactions with the enthusiastic peers I met during my tenure as secretary.

It is very important to have a group of peers with whom you can discuss, debate, and share your opinions and interests, to learn from, and to teach. Nowadays it's not that difficult or uncommon to have a whole conversation with an AI agent and however informed it may be, what it lacks is the "crazy". Humans have the capability to not follow patterns. We can go wild with our thoughts and some of the greatest discoveries and inventions in history have been born from colouring outside the lines.

The foundation of research of any magnitude lies in its effective communication. I feel PSIT has facilitated the communication of interesting Physics to people from all walks of life, and helped enthusiastic learners find a conducive space to bloom and will continue to do so. I am confident of the further progress of the club as we have great mentors, Dr Soumen Basak and Dr Suraj Hegde, with us and a highly competent and zealous team for the upcoming tenure.

I am very thankful to Dr Vinayak B. Kamble, who has guided and inspired our team to take up new challenges and do the best for the student community. I highly appreciate that many of our members are continuing to stay with the club which is a testament of their involvement and commitment. Last but not least, I am very grateful to all the team members who joined us, brainstormed new ideas and brought them to life in the best way possible.

Hope you enjoy reading our magazine and join us in trying to "hear the music".





MENTOR'S ENTANGLED ECHOES

DR. VINAYAK KAMBLE
Former FIC

Dear PSITians,

It has been a long and truly rewarding journey with PSI(t) so far. Since its inception in 2019, I have been associated with the young turks of the club as a faculty coordinator. The student volunteers kept changing every year, but I remained stationary, and now possibly a little younger than a dinosaur. Over time, I had the pleasure of working with so many of you - Ravikiran, Aadi, Saswat, Adam, Jacob, Gowri, Kaviranjana, Ishani, Harshdeep, Patatra, Krtin, and many others.

PSI(t) has had an analytical continuation from its predecessor, Anvesha, the informal science club where physics enthusiasts of the institute once gathered. The foundations of PSI(t) were partly laid through those early activities, involving students like Anand EP, Abhay Hegde, Chithran, Satyam Sahu, Arun Sathyan, and many others whom I still remember vividly. I was fortunate to coordinate Anvesha as well. Many of PSI(t)'s current activities - be it quizzes, Aficionados, or peer discussions (erstwhile journal clubs) - find their roots in those early events. As the institute moved to the new campus and our student strength grew exponentially, the physics enthusiasts finally came together as a formal club. Your super super seniors creatively named it the Physics Society of IISER TVM, cleverly abbreviated as PSI(t) - $\psi(t)$! What better symbol for a physics club than the time-dependent wave-function? Indeed, the club has been a time-evolving wave-function, and I have had the joy of being a passive observer in its evolution (I tried my best not to collapse the system!).

Over the past six years, the club has seen many milestones and initiatives - celebrating its foundation week with great enthusiasm, inviting speakers from across India and abroad (virtually too), supporting the department during the Frontiers symposium, engaging in school outreach, developing demos for Crucible, and above all, nurturing

the institute's community of physics lovers. Through it all, many of you have found creative and inspiring ways to learn and share the joy of physics. The club's social media handles are a testament to this creativity, especially the Instagram page, where one can find an array of stunning Sci-art works and event posters.

This brings me to a question: What does PSI(t) offer its volunteers beyond just doing physics?

Of course, there's more physics - different, exciting, peer-driven - but also much more. Volunteers gain skills in teamwork, communication, leadership, public speaking, writing, event planning, fundraising, and accountability - things that the classroom alone rarely teaches. I'm sure many PSITians would agree that they now feel more confident and better equipped with essential soft skills. We've come a long way - from a club once run exclusively by BS-MS students to one that now includes core members from MSc, iPhD, and even the rarely visible PhD students (somehow, we managed to drag them in!). And from leadership being mostly male-dominated, we've seen inspiring women like Ishani, Anuthi, and Khushi take up major responsibilities. I hope to see a woman president of PSI(t) soon - perhaps even from programs beyond BS-MS.

As I write this note, I would like to let you know that I will be handing over the baton of PSI(t) coordinatorship to another colleague - likely younger, and hopefully more in resonance with your energy and enthusiasm. Rest assured, I will continue to be around, ready to join, support, and advise whenever needed. Thank you, PSITians, for the wonderful journey. It has been a privilege to walk alongside you all.



EDITOR'S WAVEFRONT

PRAGUN NEPAL

Editor-in-Chief, BSMS Batch '23

Welcome to the 1st issue of Entangled, the PSI(t) magazine!

This magazine is a snapshot of what makes PSI(t) alive. Not just the finished results, but the process that leads there: curiosity chased for its own sake, the messy excitement of a lab that didn't go to plan, the quiet satisfaction of a derivation finally falling into place, and the playful art that helps us see physics in new colours. It is about the moments in between lectures and exams - when questions linger, ideas collide, and learning becomes personal.

Within these pages, you'll find work written by volunteers from across batches and backgrounds. Some pieces aim to simplify difficult ideas without diluting them. Others document projects that may still be unfinished, but are honest records of effort and learning. There are outreach stories that remind us why we share science beyond our classrooms, and attempts at science communication that try to make complicated concepts feel tangible - something you can hold in your hand, turn around, and look at from different angles. Together, they reflect the many ways one can engage with physics.

As Editor-in-Chief, my goal for this first issue was deliberately simple: to keep the magazine readable, honest, and useful. Readable meant clarity - no unnecessary jargon, and explanations that respect the reader's time and curiosity. Honest meant allowing authors to talk about confusion, false starts, and imperfect outcomes, because that is how real science unfolds. Useful meant creating something that could serve multiple purposes: a reference to return to, a spark for a new idea, or a starting point for a conversation.

A magazine, however, is only meaningful if it sparks something in the reader. Read

this issue in whatever way suits you: as a reference, as inspiration, or simply as something to flip through between classes. If an article makes you curious, follow that curiosity beyond the page. If something frustrates you or feels incomplete, take that as an invitation to ask a question, start a discussion, or begin a project of your own. In clubs like ours, the most valuable outcomes are often the small, informal conversations that slowly grow into collaborations.

This issue exists because of collective effort. My deepest thanks to everyone who wrote, edited, illustrated, and supported Entangled. To the volunteers who polished drafts late into the night, the seniors who offered patient guidance, our faculty advisors who backed our events and ideas, and the readers - whose curiosity and feedback keep us honest and motivated. This magazine is as much yours as it is ours.

If something in this issue sparks a thought, I hope you'll reach out. Pitch a piece, suggest a theme, question an argument, or join an event. We are always looking for more voices, more perspectives, and more experiments - both literal and figurative. Entangled is meant to be a curious space: a place where ideas interact, interfere, and occasionally produce something unexpectedly beautiful.

This first issue is only a beginning. Think of it as an evolving wave-function; shaped by every question asked, every contribution made, and every conversation that follows!



1

THE SPIRIT OF ENTANGLED

WHERE IDEAS INTERSECT: THE SPIRIT OF ENTANGLED

Physics has a long history of synthesizing complex phenomena into a few unifying theories. With the rapid revolution in science and technology, physical science has emerged as an ultimate challenge in the quest to discover the truth about our world. From its fundamental principles, physics seeks to explain how matter and energy interact through observation, experimentation, and mathematical reasoning.

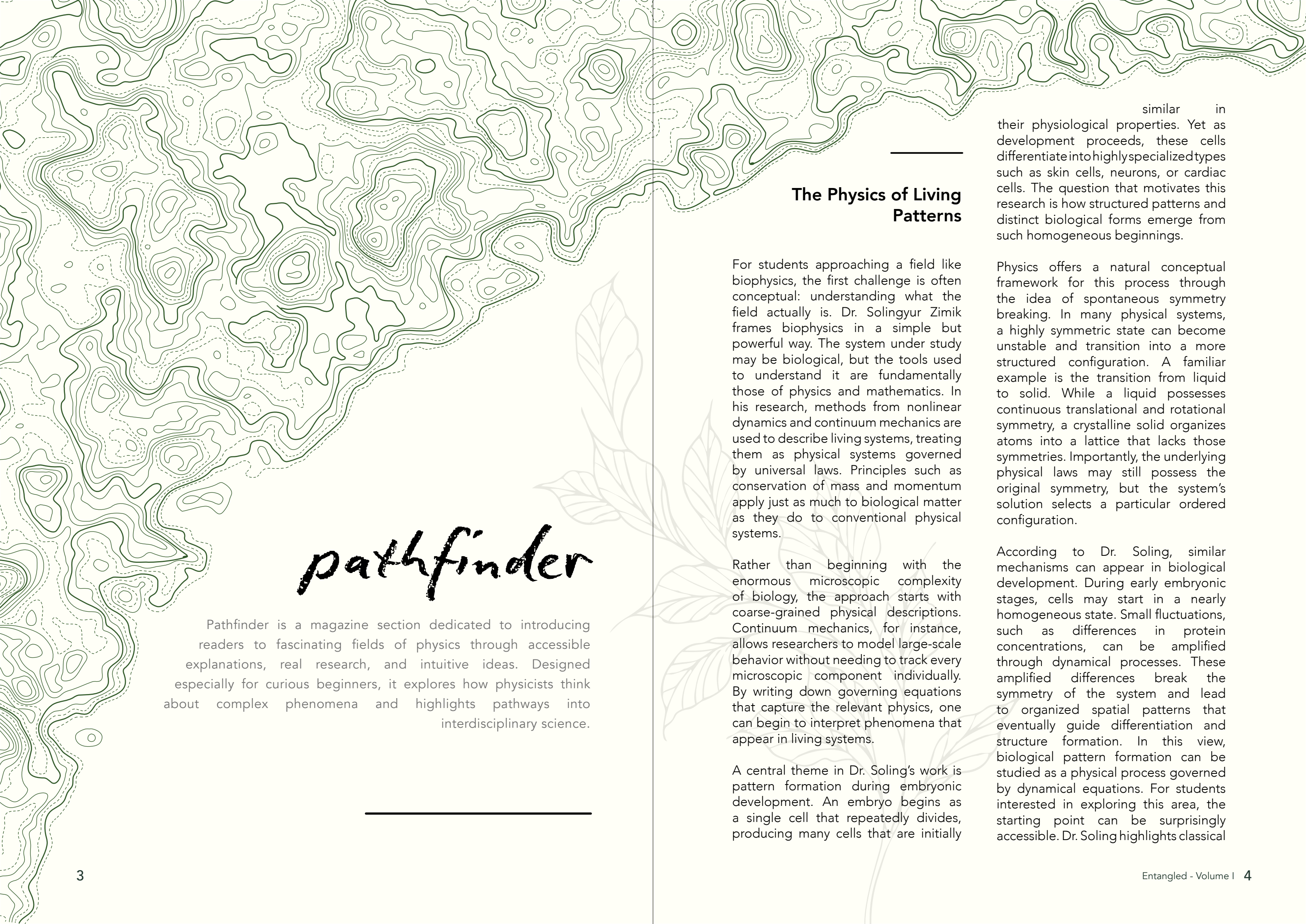
The mysterious and non-intuitive beauty of physics connects ideas from the classical to the modern era, revealing the interdependence of physical laws and the phenomena surrounding us. From mechanics, thermodynamics, electromagnetism, and optics to modern quantum phenomenology, the study of physics provides a framework for understanding the diverse and fascinating phenomena of the natural world. Over time, physics has developed as **an ever-growing network of ideas**. With Albert Einstein's theory of relativity, new insights were offered into the cosmic framework of space, time, and gravity. Through the pioneering efforts of scientists like Planck, Bohr, Heisenberg, and Schrödinger, the behavior of particles at the atomic and subatomic levels was explained through quantum mechanics. Nuclear physics, with applications ranging from energy production and astronomical studies to national defense, became highly significant. Astrophysical mysteries began to be solved: from the Big Bang theory to the discoveries of dark matter and dark energy, these breakthroughs have revolutionized our understanding of the cosmos. **None of these discoveries stand in isolation** - each leads to another. Newtonian mechanics describes the motion of objects and the forces between them, while Maxwell's equations describe the behavior of electromagnetic fields. The apparent conflict between the two emerged because Maxwell's equations imply a constant speed of light, which contradicts the Galilean transformations underlying Newtonian mechanics. This conflict led to the development of Einstein's theory of special relativity, which reconciled the two by showing that space and time are relative, not absolute, and that the speed of light is constant for all observers. Quantum entanglement is another remarkable phenomenon in quantum mechanics, where two or more particles are linked in such a way that the state of one instantly influences the state of the other, no matter how far apart they are. **To unveil the mysteries of nature, every branch of physics provides solutions that remain fundamentally entangled with one another.**

Modern physics does not stand apart from the rest of science. Interdisciplinary studies are rapidly evolving, bridging the fundamental principles of physics with other scientific disciplines. Through rigorous methods and quantitative models applied to diverse and complex domains, physics has driven major breakthroughs in fields such as biophysics, materials science, medical imaging, environmental science, and

nanotechnology. For example, chaos theory helps ecologists understand unpredictable population dynamics - such as sudden booms, crashes, and irregular predator-prey cycles - thereby illuminating the stability, resilience, and extinction risks within ecosystems. Quantum computing harnesses the principles of quantum mechanics to tackle problems that are computationally intractable for classical computers, promising to revolutionize chemistry and condensed matter physics. From microscopic behavior to macroscopic properties, statistical mechanics is vital for understanding large collections of biomolecules, bridging physics with the life sciences. Interdisciplinary physics not only deepens our understanding of the natural world but also drives technological advancements that address real-world challenges. **Physics remains deeply entangled with the whole of science.**

Science is social; discovery is a collective act. It never grows in isolation. It thrives on open discussion, constructive debate, and shared knowledge through mentorship. Classrooms and research labs are spaces where ideas are challenged, refined, and strengthened over time. Like particles in quantum entanglement, students are connected through their learning journeys and shared experiences. Every question asked invites collaboration, making our knowledge richer. We shape one another's understanding, building a **network of knowledge that grows stronger through connection**. Our dynamic club brings together curious minds united by a shared passion for physics. The intellectual network we foster becomes a source of fresh perspectives and unexplored ideas, where everyone's voice adds to a larger one. With this magazine, we weave diverse reflections into its pages, encouraging readers to think deeply and explore how physics helps us understand our universe and how it affects our communities and environments. It is a step towards fostering a new generation of scientists. This magazine speaks to anyone who is curious, whether they are seasoned physicists or newcomers with wide-eyed questions. It invites readers to join the conversation, ask bold questions, and share in the joy of research and discovery. **Learning physics is not a solitary trek but a communal adventure.** It allows us to come together to wonder aloud and puzzle over scientific problems, creating a vibrant atmosphere of curiosity. This shared exploration makes complex ideas accessible, memorable, and alive in ways that solitary study rarely can. Through your journey in this magazine, you become part of a living, entangled web of curiosity and ideas, one that grows with every page you turn. **With "Entangled", This connection should not end when you reach the last page. Instead, it should spark your thoughts and inspire you to reach outward, continuing the shared journey of exploring the science that ultimately shapes our world.**

- Anay Mishra, BSMS'24



pathfinder

Pathfinder is a magazine section dedicated to introducing readers to fascinating fields of physics through accessible explanations, real research, and intuitive ideas. Designed especially for curious beginners, it explores how physicists think about complex phenomena and highlights pathways into interdisciplinary science.

The Physics of Living Patterns

For students approaching a field like biophysics, the first challenge is often conceptual: understanding what the field actually is. Dr. Solingur Zimik frames biophysics in a simple but powerful way. The system under study may be biological, but the tools used to understand it are fundamentally those of physics and mathematics. In his research, methods from nonlinear dynamics and continuum mechanics are used to describe living systems, treating them as physical systems governed by universal laws. Principles such as conservation of mass and momentum apply just as much to biological matter as they do to conventional physical systems.

Rather than beginning with the enormous microscopic complexity of biology, the approach starts with coarse-grained physical descriptions. Continuum mechanics, for instance, allows researchers to model large-scale behavior without needing to track every microscopic component individually. By writing down governing equations that capture the relevant physics, one can begin to interpret phenomena that appear in living systems.

A central theme in Dr. Soling's work is pattern formation during embryonic development. An embryo begins as a single cell that repeatedly divides, producing many cells that are initially

similar in their physiological properties. Yet as development proceeds, these cells differentiate into highly specialized types such as skin cells, neurons, or cardiac cells. The question that motivates this research is how structured patterns and distinct biological forms emerge from such homogeneous beginnings.

Physics offers a natural conceptual framework for this process through the idea of spontaneous symmetry breaking. In many physical systems, a highly symmetric state can become unstable and transition into a more structured configuration. A familiar example is the transition from liquid to solid. While a liquid possesses continuous translational and rotational symmetry, a crystalline solid organizes atoms into a lattice that lacks those symmetries. Importantly, the underlying physical laws may still possess the original symmetry, but the system's solution selects a particular ordered configuration.

According to Dr. Soling, similar mechanisms can appear in biological development. During early embryonic stages, cells may start in a nearly homogeneous state. Small fluctuations, such as differences in protein concentrations, can be amplified through dynamical processes. These amplified differences break the symmetry of the system and lead to organized spatial patterns that eventually guide differentiation and structure formation. In this view, biological pattern formation can be studied as a physical process governed by dynamical equations. For students interested in exploring this area, the starting point can be surprisingly accessible. Dr. Soling highlights classical

mathematical models that illustrate how patterns emerge from simple rules. One well-known example is the Turing model, which describes how interacting chemical species diffusing through space can spontaneously generate patterns such as stripes or spots, similar to those seen on animal coats. These models are expressed through partial differential equations and can be studied by anyone with a background in calculus and differential equations.



Dr Solingur Zimik Kachui

Another useful starting framework comes from phase separation in binary mixtures. When two components are mixed at high temperature, they may form a uniform state. As the system cools, differences in interaction strengths can cause the components to separate into distinct regions. This basic physical mechanism provides an intuitive analogy for how cellular components might segregate during developmental processes, offering a simple model for studying pattern formation.

Dr. Soling also points out that interdisciplinary work of this kind has not always been common. Historically, physicists, mathematicians, and biologists tended to work within separate intellectual boundaries. Only in recent decades have these boundaries begun to blur, allowing



Microrobots for pulmonary drug delivery
Courtesy: [Zhengxing Li et al. at nature.com](#)

quantitative approaches to address biological questions. In many research environments, especially outside major institutions, this integration is still developing, which makes the field both challenging and promising.

Despite these challenges, the opportunities are substantial. Biological systems contain many unresolved problems that can be approached with relatively modest mathematical tools combined with computational methods. For instance, Dr. Soling studies electrical wave propagation in the heart, where certain spatial and temporal patterns are associated with irregular cardiac rhythms. Understanding how such patterns form and how they might be controlled could have meaningful medical implications.

Taken together, his perspective offers a clear pathway for students: build a strong foundation in physics and mathematics, explore simple dynamical models of pattern formation, and gradually extend those ideas to complex biological systems. Through this approach, the physics of living matter becomes not an intimidating frontier, but a domain where quantitative thinking can uncover the principles behind life's intricate structures.

- Shamayela Fayaz MSC'24

Rethinking Scientific Communication

Khushi Tiwari (BSMS'22)

Science has long been celebrated as a universal language; a shared pursuit that crosses borders, cultures, and centuries. Ideas in physics have traveled across continents, translated into new tongues, reshaped through collaboration, and built upon by generations. The core strength of science lies in its openness and in its ability to evolve through the free exchange of ideas.

But while scientific thought may be universal, scientific communication is not. The ways we talk about science, through lectures, papers, equations, and diagrams, rely heavily on specific forms of language: spoken, written, and visual. And these forms, however powerful, are not accessible to everyone.

For Deaf and hard-of-hearing learners, physics education often centers on forms of communication that are inaccessible by design. Spoken lectures, rapid verbal explanations, and audio-heavy content dominate many classrooms, with little space made for sign language, visual timing, or the pace of interpretation. Written language, while helpful, is not always a complete substitute; it can carry assumptions, jargon, and missing context that are harder to grasp without direct discussion in one's primary language. Even though the ideas of

science move freely, the ways we deliver those ideas can unintentionally create walls. And for many learners, those walls aren't about ability, but about access.

Science thrives on diversity, not just of disciplines but of perspectives, lived experiences, and modes of understanding. Throughout history, breakthroughs have often followed representation. For instance, women's health research remained underfunded and understudied until more women became scientists and shifted the research agenda to include



Courtesy: [Sasan Rashtipour](#) on Unsplash

neglected issues. Inclusion isn't a matter of charity; it's a catalyst for discovery. The same is true for Deaf and hard-of-hearing (DHH) communities. According to the 2011 Indian census, India is home to over 50 lakh (5 million+) Deaf or hard-of-hearing individuals. And yet, science remains a field that communicates primarily

through speech and text, often excluding learners who experience the world differently.

While progress has begun, for example, NCERT science textbooks for a few grades are now available in Indian Sign Language (ISL), systemic barriers persist. Most deaf students



Indian Sign Language (ISL)

still struggle to find schools that offer the science stream at the higher secondary level, not due to a lack of interest but due to a lack of trained teachers, accessible materials, and institutional support. This linguistic gap limits both learning and participation. Without a rich scientific vocabulary in ISL, deaf students face unnecessary barriers to becoming scientists, researchers, or even curious citizens of the scientific world.

So, what would a more inclusive physics look like? Could we imagine physics taught not just through speech or text but through sign language, tactile tools, gesture, sound,

and embodied learning?

Initiatives like CSIR's JIGYASA-ISL EVL program are working to bridge the divide by co-developing science education content in Indian Sign Language. But this can't be a top-down solution. Building a truly inclusive scientific culture requires collective effort from scientists, educators, designers, and policy-makers. Indian Sign Language (ISL) has evolved over time into a powerful, expressive, and structured language with its own grammar and visual logic. Far from being a simplified tool, ISL is rich in nuance and capable of conveying complex ideas, including those from science and mathematics. Organizations like the Indian Sign Language Research and Training Centre (ISL-RTC) have taken significant steps in standardizing and promoting ISL, including creating a growing online ISL dictionary and subject-specific glossaries, such as in physics and biology. Their publicly available website is a valuable resource and a testament to the importance of linguistic inclusivity in education. I will be attaching the link to this resource, along with further material from the ISL EVL initiative, for reference and to support continued exploration and engagement.

These institutional efforts are deeply admirable, not just because they address a long-overdue gap, but because they model what inclusive infrastructure can look like. Yet, this vital work is only the beginning. Widespread accessibility in science will require broader community participa-

tion. That's where colleges, students, professors, and academic groups can play a transformative role. Let's look at what we can do next.

- Students, professors, and college clubs can play a powerful role in bridging the gap between science and sign language. One of the most meaningful actions is to collaborate with the Deaf community and ISL experts to help co-create signs for scientific terms that currently don't exist. This ensures the vocabulary is both linguistically accurate and culturally respectful, rather than imposed from the outside.
- Student-led initiatives can start documenting scientific concepts through short videos where Deaf and hearing students demonstrate or explain terms in Indian Sign Language. These small, crowd-sourced projects can grow into open-access visual dictionaries supporting classrooms nationwide. Clubs can host events where science is taught or demonstrated in sign language, raising awareness and normalizing ISL in academic spaces.
- Emerging technologies like AR, VR, and MR open new frontiers in accessible science education. For Deaf and hard-of-hearing students, immersive visual environments can offer intuitive, spatial ways to experience physical concepts. Sign language can also be embedded directly into virtual learning environments, enabling

real-time, gesture-based guidance from avatars or interpreters. MR tools, which blend physical and digital spaces, can allow learners to move through simulations using body gestures, making physics feel tactile, embodied, and inclusive. These approaches support translation and transformation, reshaping how physics is taught, felt, and understood. While some of these tools are still in early development, experimental classroom trials and accessibility hackathons have already begun exploring these possibilities. Integrating extended reality into inclusive design can help create multisensory, multimodal learning spaces where communication isn't limited to words or chalkboards, but becomes dynamic, visual, and interactive.

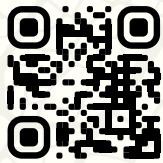


Baidu XiLing Sign Language Digital AI Avatar used in 2022 Winter Paralympics Games
Courtesy: PR newswire

- Professors can include accessibility-focused projects in their syllabi, encouraging students to think about inclusive education not as a separate field but as an essential part of scientific communication. Introducing basic ISL training in science and education courses helps build a foundation for more

inclusive communication in physics and beyond. By embedding these priorities into mainstream education and research, college communities can help create a version of science that speaks visually, spatially, and inclusively to everyone.

Deaf scientists, queer scientists, women scientists, and disabled scientists don't just "benefit" from inclusion, they transform the questions we ask and the tools we build. Inclusion isn't just about access but about changing who gets to shape science. When we broaden the ways physics is taught and communicated, we invite not just more learners into the classroom but more ideas into the field. Building a truly inclusive physics community means embracing diverse ways of knowing and communicating. By working together to expand scientific vocabulary in Indian Sign Language and integrating sign language into education, we open doors for countless learners to engage, contribute, and innovate. Inclusion isn't just an add-on; it's essential for the future of science itself.



Visit the Indian Sign Language Virtual Lab

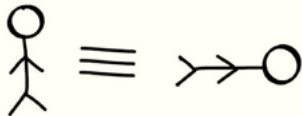
Psitons

Feel a pull towards your phone?



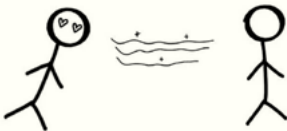
Gravity

Clumsy?



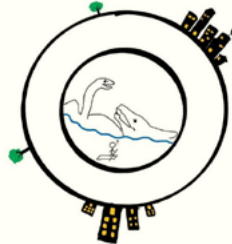
Gravity

Attracted to another?



Gravity ?

Journey to the centre of the Earth?



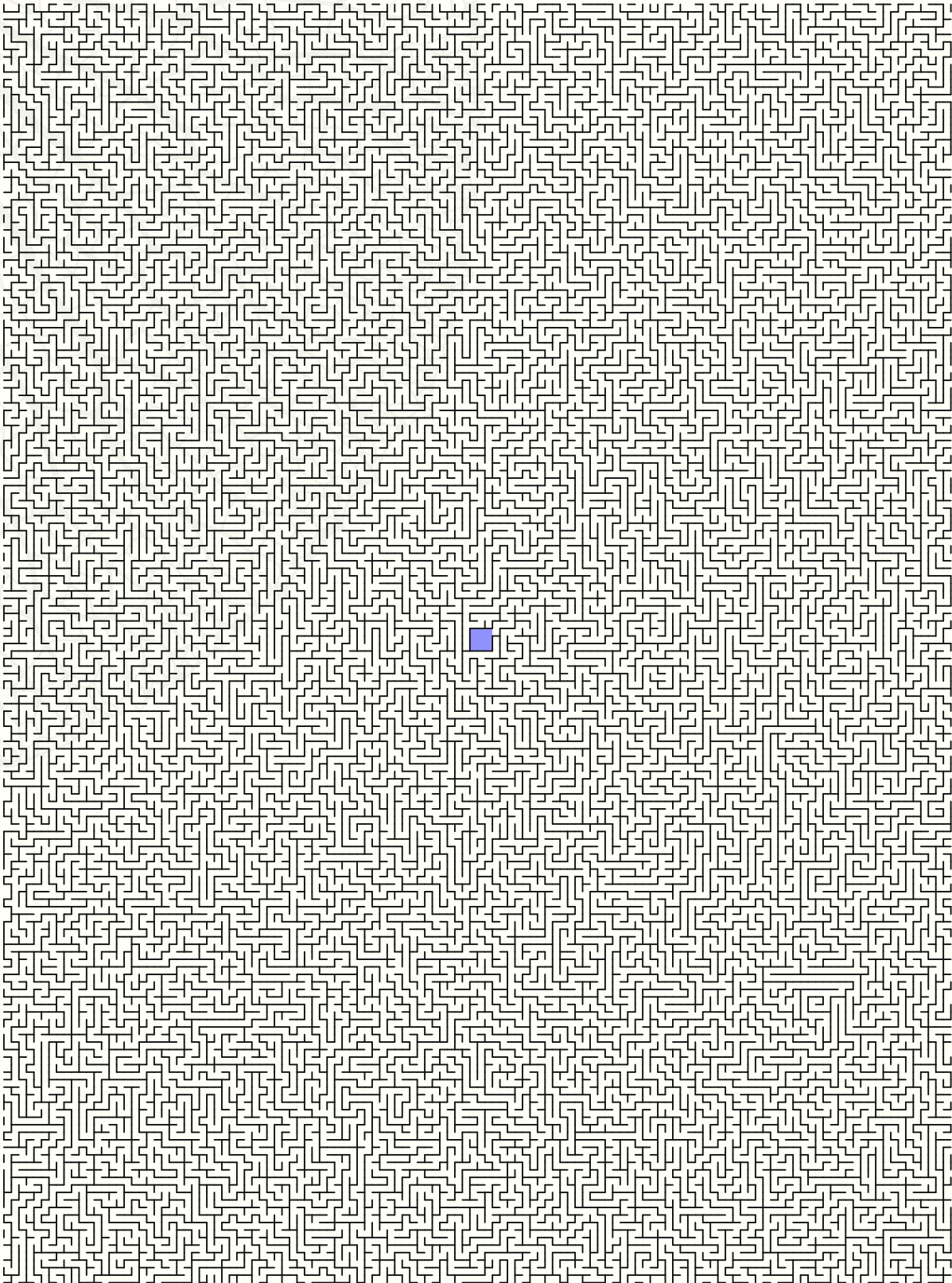
Don't blame gravity

Be sure to get the stickers!
We sell them every once in a while!



the LABYRINTH

Determine a deterministic path from the coordinate center to the system boundaries.



Solution on page 20. Challenge yourself before looking!

FictionInAction

When Storytelling Becomes the First Law of Discovery

Eldho Sam Varghese (BSMS'22)

We've all been there. It's two in the morning, the faint glow of a screen illuminating a pile of snack wrappers, and you've just watched a starship captain calmly order a jump to warp speed. A part of your brain, the logical one, dismisses it with a comfortable sigh: "Cool, but that's just make-believe." But is it? Is the line between the speculative page-turner

and the peer-reviewed paper truly so absolute? What if the wildest dreams of our most imaginative writers are not mere fantasy, but are, in fact, the blueprints for our reality? For centuries, science fiction has served as humanity's thought experiment, a grand cultural sandbox in which we can play with the universe's most profound and terrifying "what ifs"

The Shadow of M87: * Captured by the Event Horizon Telescope, this image shows the radio emission from superheated gas swirling around a supermassive black hole. The central dark region is the "shadow" cast by the event horizon, a physical manifestation of the spacetime collapse first predicted by Schwarzschild in 1916.

without blowing up the lab. It is the realm of the adjacent possible, a place to explore the consequences of ideas just beyond our technological grasp. But every so often, the barrier between that sandbox and the real world becomes porous. An idea, born in the mind of a storyteller, seeps out. It might infect a young physicist with a lifelong passion, or provide a sudden, shocking insight to a scientist wrestling with a paradox. This is the tale of that leakage. It's the story of how poetry preceded physics, how nightmares became blueprints, and how the quest to build a prop for a movie ended up changing our understanding of the cosmos. Buckle up, because we're diving into the uncanny valley where fiction stops being fiction and starts making history.

The universe is vast and silent, but our stories have always filled it with wonders and monsters. Well before we possessed telescopes that could pierce the veil of deep space, our imaginations were already populating the darkness. Picture the 1830s. Science is a world of clockwork precision, of Newtonian laws governing the elegant dance of planets. Into this world of light and order steps Edgar Allan Poe, the master of the macabre. In his 1835 tale, "The Unparalleled Adventure of One Hans Pfaall," his narrator glimpses a chilling anomaly on his journey to the moon: a "dark star," a region of space described as a dead sun, whose gravitational pull is so absolute, so complete, that not a spark of light could escape its hold. It is not merely dark; it is a void, a patch of total nothing-

ness against the tapestry of the heavens. It's a gothic concept in its purest form, a haunted house on a cosmic scale. Sixty years later, H.G. Wells, a writer with a more scientific bent,

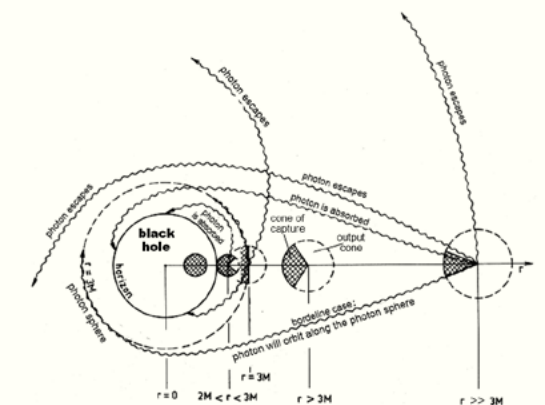


Illustration of the Schwarzschild Radius

explored a similar idea in his 1897 story, "The Star." A mysterious celestial body wanders into our solar system, not as a brilliant comet, but as an invisible phantom. Its presence is felt, not seen. It yanks planets from their orbits, its immense gravity acting like an unseen hand stirring the celestial pot. Wells's "dark star" was a cosmic bully, an entity defined not by what it was, but by the gravitational chaos it caused. Both Poe and Wells were describing, with stunning intuition, the object we now know as a black hole. Their concepts, at the time, were mere fantasy. It wasn't until 1916, equipped with Albert Einstein's groundbreaking new theory of general relativity, that the German physicist Karl Schwarzschild sat down to solve its complex equations for a single, non-rotating spherical mass. To his surprise, the mathematics unveiled a breaking point. If sufficient mass were packed into a small

enough space, the fabric of spacetime itself would collapse, producing an

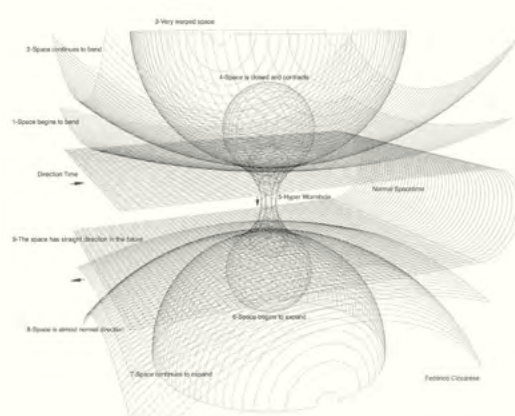


NGC 6302, as taken by Hubble Space Telescope

object with an escape velocity greater than the speed of light. He had discovered Poe's dark star, Wells's gravitational phantom. It was a singularity in spacetime, encased in an event horizon - a one-way membrane from which nothing could return. The catchy title "black hole" wouldn't be coined by physicist John Wheeler until 1967, but the beast had already been lurking in our fiction for more than a century. It was an astonishing instance of parallel imagination, where the human mind, contemplating infinity and nothingness, arrived at the same frightful conclusion as the most sophisticated mathematics.

If black holes are the dead ends of the universe, wormholes are its secret passages. The dream of a shortcut across the cosmos is a staple of science fiction, a necessary storytelling device to prevent tales in which char-

acters expire of old age before they reach the next star system. In 1865, Lewis Carroll showed us a fantastical, frightening version with Alice's Adventures in Wonderland. Alice's tumble down the rabbit hole is more than a mere descent; it's a dizzying trip through a surreal tunnel that defies the regular laws of space and time, linking a tranquil English riverbank to the mad realm of Wonderland. Almost a hundred years later, Madeleine L'Engle gave the idea a scientific-sounding veneer in her beloved 1962 novel, *A Wrinkle in Time*. The book's otherworldly guides describe the "tesseract" not as a tunnel, but as a fold. Visualize an ant on a long piece of string. To travel from one side to the other, it has to crawl the entire length. But if you fold the string, bringing both ends together, the ant can cross in an instant. L'Engle's tesseract was a means of "wrin-



Schematic of a theoretical wormhole.
Courtesy: Federico Ciccarese (CC BY-SA 4.0)

klung" spacetime, a shortcut through the fifth dimension. It all seemed like lovely, imaginative fantasy. Only, it wasn't. In 1935, Albert Einstein and his collaborator Nathan Rosen were studying the strange implications

of general relativity and discovered that their equations permitted the existence of "bridges" spanning two distant points in spacetime. These



Prof. Kip Thorne's Renaissance moment
Courtesy: Alexander Mahmoud

Einstein-Rosen bridges were, for all practical purposes, mathematical wormholes. For decades, they remained a curiosity, a mathematical oddity with no discernible practical application. Then, in the 1980s, science fiction stepped in directly. The astronomer Carl Sagan was writing his novel *Contact* and required a scientifically realistic means for his heroine, Dr. Ellie Arroway, to travel to the star Vega. He initially pictured her plummeting through a black hole, but he knew that wasn't realistic. So, he did what any good scientist does: he went to an expert. He phoned his friend, the renowned theoretical physicist Kip Thorne. Thorne realized that a simple black hole would not do - it would spaghetti Arroway into a stream of atoms. But the problem intrigued him. Could a wormhole, an Einstein-Rosen bridge, be made traversable? Thorne brought the problem back to his Caltech students. Together, they worked through the math.

They found that a wormhole would collapse instantly unless it was supported open by an exotic substance with negative energy density - something they called "exotic matter." This research, commissioned by the plot needs of a science fiction novel, led to several seminal scientific articles on the physics of traversable wormholes. It was a direct, traceable pathway from fiction to fundamental theory.

While some writers looked to the stars, others gazed inward, to the atom, and discovered in its tiny core a power both divine and demonic. In 1914, as the world careened toward the industrial carnage of World War I, H.G. Wells wrote a novel that was at once a scientific marvel and a chilling prophecy. In *The World Set Free*, he foresaw a future war not fought with machine guns and cannon, but with a fearsome new technology: "atomic bombs." Wells, who had followed with enthusiasm the early findings of radioactivity, envisioned that humanity might harness the power in the atom. His bombs weren't merely bigger explosions; they released a "continual explosion," a chain reaction that would burn for days, reducing cities not only to rubble, but radiating an invisible, lethal poison. At the time, the notion was viewed by most as fantasy. The world's most prominent scientists, including Ernest Rutherford and even Einstein, thought it impossible to harness atomic power on a mass scale. Then, in 1933, a restless but brilliant Hungarian physicist, Leo Szilard, was strolling through the streets of Lon-



The Manhattan Project squad: Bringing H.G. Wells's atomic nightmares to life in 1945.
Courtesy: Science History Images

don, contemplating the future. He had just finished reading *The World Set Free*. As he stood at a traffic light, waiting to cross the street, it all fell into place in his mind: Wells's chain reaction. If a neutron could strike an atom and release two more neutrons, which themselves could strike two more atoms, the process would spread exponentially, releasing catastrophic amounts of energy. The concept, which began as a product of science fiction, became his obsession. This very idea, conceived at a London street corner and sparked by a novel, became the scientific basis

for the Manhattan Project. Szilard himself would proceed to write the infamous letter, signed by Einstein, warning President Roosevelt of the threat of a German atomic bomb, which put into motion the race to produce the very weapon Wells had imagined. It is one of history's great ironies: a cautionary tale that directly inspired the invention of the technology it cautioned against.

Turn on an episode of *Star Trek: The Next Generation*, and it is certain you will hear the chief engineer report on the status of the "matter-antimatter

reaction chamber." The iconic warp core, with its glowing blue light, runs on the most efficient and powerful energy source conceivable: the controlled annihilation of matter and its mirror-image, antimatter. For many generations of fans, this was the pinnacle of clean, futuristic energy. The idea of antimatter, however, didn't come from a writer's room. It came from the blackboard of one of the 20th century's quietest geniuses, Paul Dirac. In 1928, Dirac was trying to solve a major problem in physics: how to create an equation for the electron that was consistent with both quantum mechanics and Einstein's special relativity. The lovely equation he derived was perfect, but it contained a peculiar anomaly. Just as the equation $x^2 = 4$ has two solutions (2 and -2), Dirac's equation had two solutions. One described the ordinary, negatively charged electron. The other described a particle of the same mass but with a positive charge - an "anti-electron." Initially, nobody knew what to make of this mathematical ghost. Dirac himself was uncertain. But he believed his math and foresaw that such a particle had to exist. A couple of years later, in 1932, physicist Carl Anderson was researching cosmic rays and discovered tracks in his cloud chamber left by a particle with the exact characteristics of Dirac's anti-electron. He called it the "positron." Antimatter existed. While science discovered its existence, it was science fiction, especially *Star Trek*, that imagined its ultimate application. The show took an esoteric particle, known only to physicists, and transformed it

into the cultural touchstone for ultimate power, inspiring generations to dream of harnessing the universe's fundamental symmetry.

Sometimes, fiction doesn't just predict a discovery; it provides the blueprint for an invention, a clear and compelling vision that engineers then strive to make real. Arthur C. Clarke was a science fiction giant, but he was also an extraordinary engineer with a love for radar and communications. In October 1945, many years before the first satellite had even been put into orbit, he wrote an article in a small magazine called *Wireless World*. The article, entitled "Extra-Terrestrial Relays," was not a story but a technical proposal. Clarke worked out that if a satellite were positioned in an orbit 22,236 miles from the Earth's equator, its orbital period would be exactly 24 hours. From the ground, it would seem to hang motionless in the same location in the sky. He then sketched out the vision: a network of just three of these "geostationary" satellites, evenly spaced around the world, could provide television and telephony to the entire planet. It was a complete, elegant, and fully imagined plan for our global communications network. It was an audacious fantasy at the time. But in 1964, NASA launched Syncom 3, the first geostationary communications satellite, which relayed the Tokyo Olympics to America. Clarke's vision became the foundation of our contemporary world. In honor of his foresight, the geostationary orbit is now officially called the Clarke Orbit.

In Neal Stephenson's 1995 cyberpunk classic, *The Diamond Age*, society is centered around "Matter Compilers." These are wonderful machines, found in every home, that can build nearly anything - food, clothing, tools, even weapons - by arranging individual atoms according to a digital plan. It is the ultimate post-scarcity technology, a dream of creation at the most basic level. This vision, however, was first expressed not by a fiction writer but by a physicist with a notoriously creative mind: Richard Feynman.



Little Brother is watching us
Courtesy: Britannica

In 1959, in a talk at Caltech entitled "There's Plenty of Room at the Bottom," Feynman challenged his fellow scientists. He posed, "Why can't we write the whole 24 volumes of the *Encyclopædia Britannica* on the head of a pin?" He then explained, step by step from fundamental principles, why the laws of physics did not prohibit the control of single atoms. We could, he claimed, construct tiny machines that would be capable of constructing still smaller machines, right down to the size of an atom. He was

suggesting a whole new area of engineering. Feynman gave the scientific challenge, the assertion that the territory was available for exploration. But it was science fiction like Stephenson's that vividly described what that territory might be like once inhabited. Authors provided us with the vision of what nanotechnology could be when matured, filling the public imagination and inspiring a new breed of scientists, such as Eric Drexler, who would go on to codify the principles of molecular engineering.

Few images in science fiction are as iconic as the shimmering distortion in the air when a Romulan Bird-of-Prey activates its cloaking device in *Star Trek*. The ship doesn't vanish; it warps light around itself, rendering it invisible to the eye and to detectors. It was a dazzling bit of television wizardry, the ultimate stealth and surprise device. For decades, it was strictly in the realm of fantasy. Vision is possible because light reflects off an object and comes to our eyes. If you wanted to be invisible, you would need to somehow prevent this from occurring. Scientists, explicitly referring to *Star Trek* and other science fiction, started to question in the early 2000s whether it was really impossible. A group led by Sir John Pendry suggested the idea of "metamaterials." These are not naturally occurring materials but man-made structures with minuscule patterns that can play with electromagnetic waves in outlandish ways. Pendry hypothesized that it would be possible to design a shell of metamaterials



These **Four** look **Fantastic**
Courtesy: Artemis II, NASA

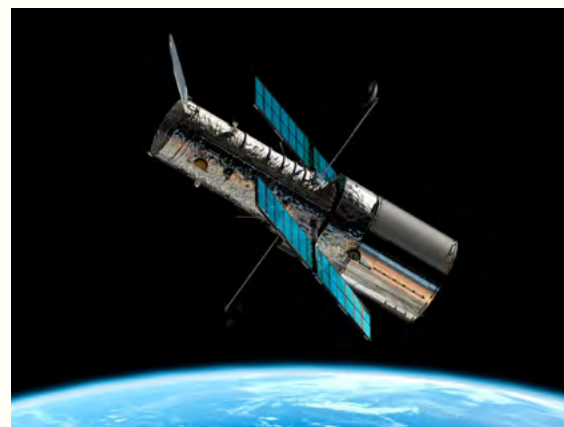
that would guide light waves around an object, such as water around a boulder, so that they emerge on the other side as if they had traveled through empty space. In 2006, that hypothesis was tested. A group of scientists successfully cloaked a tiny copper cylinder from microwaves. Though a human-sized, full-spectrum cloak remains a huge challenge, the underlying principle was established. It was an explicit instance of scientists thinking, "I want to make that thing from the show," and then developing a whole new discipline of optics to accomplish it.

The deepest prophecies of science fic-

tion are frequently not about technology, but rather our comprehension of reality itself. They question our most fundamental ideas about the universe. In 1895, H.G. Wells took his nameless Time Traveler on an otherworldly adventure. Sitting in his bizarre device of ivory and quartz, he pulls a lever and observes the world outside his laboratory window speed by. The sun and moon hurtle through the sky, days and nights flashing past in seconds. Centuries pass for the world outside. For the traveler, it is just a matter of hours. Wells had instinctively written about a basic, and profoundly not self-evident, fact of the universe: time is not a fixed,

universal stream. Its passage is relative. Ten years later, a young patent clerk in Bern, Switzerland, Albert Einstein, wrote his special theory of relativity. It was a groundbreaking article that turned 200 years of physics on its head. One of its most important predictions was time dilation: the more quickly you travel in space, the more slowly you travel in time. An astronaut on a journey at close to light speed would come back to Earth having aged just a few years, when decades or centuries had gone by for everyone else. Now it is not a hypothetical idea. The GPS satellites circling Earth are traveling so quickly that their onboard atomic clocks run a fraction slower than clocks on the ground. If engineers didn't continuously make adjustments for this relativistic effect - this Wellsian insight - your GPS would be off by several miles each and every day.

What if every decision you ever made, no matter how insignificant, bifurcated the universe into two? In one, you turned left. In the other, you turned right. Both worlds are just as real, forking off into infinity. This vertiginous notion was investigated with stunning literary grace by the Argentine author Jorge Luis Borges in his 1941 short story, "The Garden of Forking Paths." The tale recounts a mythical Chinese novel that is also a maze, a work in which all possible outcomes of every occurrence exist side by side, fashioning an infinite mesh of diverging paths. This, and subsequent novels such as Philip K. Dick's *The Man in the High Castle*, that envisioned characters sliding be-



Hubble Space Telescope
Credit: NASA/ESA

tween other histories, treated the multiverse as a playground of philosophy. In 1957, a Princeton PhD student in physics named Hugh Everett III was wrestling with a profound paradox of quantum mechanics known as the "measurement problem." Particles in the quantum universe exist as a vague cloud of potentials until the moment they are measured, at which time they "collapse" into a definite state. But what triggered the collapse? Everett suggested a revolutionary, and then heterodox, answer: the collapse never occurs. Rather, each and every quantum measurement splits the entire universe. In one universe, the particle is spin-up; in another, it is spin-down. Everything that is possible occurs, in its own isolated branch of reality. Everett's Many-Worlds Interpretation was Borges's garden, transposed into the vocabulary of quantum physics. Though still controversial, the multiverse is today a standard model of contemporary cosmology, manifesting in cosmic inflation theories and string theory. Science fiction had given us the conceptual context, a manner of thinking about separate realities branching off

from one another, many years before physics required it to resolve its own most fundamental enigmas.

This eerie tango between book and physics laboratory becomes even more otherworldly when we venture into quantum. Sci-fi abounds with magical communications devices, such as Ursula K. Le Guin's "ansible," a box that allowed you to have a real-time chat with somebody on another world, with no delay at all. This clearly broke all the known laws of physics. Or so it seemed. It appears that the universe has its own ghostly equivalent. Quantum entanglement, a process Einstein himself was unable to understand, connects two particles so that they become "twins." Regardless of how far apart they are - across a room or across the universe - if you take a reading on one, the other instantly copies its state. We can't employ it to gossip throughout the universe (yet), but this rule, guessed at by fiction, is the foundation of quantum computing and teleportation - not of individuals, but of data, "beaming" the very map of a particle from one location to another. From the dark emptiness of a black hole to the endless branches of the multiverse, the pattern holds true. Science fiction is not merely escapism. It is an essential component of the scientific process itself. It is a feedback loop in which our dreams guide our discoveries, and our discoveries power increasingly audacious dreams. The process is a subtle dance. Sometimes, fiction is prophecy, an eerie vision of coming truth. Sometimes it is a direct inspi-

ration, a gauntlet thrown down to the scientists and engineers who say, "I can make that." And sometimes it is just a language, a collection of metaphors and ideas that gives us a way to speak about things so bizarre they would otherwise be unspeakable. The next great leap in our understanding of the universe may not come from a multi-billion dollar particle collider or a space telescope. It might be lurking right now, in the pages of a forgotten paperback, in a late-night movie, or in the daydreams of a storyteller. Because before we can build the future, before we can discover its laws, we must first have the courage to imagine it.

solution **the labyrinth**

If you found yourself hitting a wall at every turn, your calculations were correct. This puzzle was designed with a condition that makes an exit theoretically impossible.

In physics, as in life, we often operate under the assumption that every elegant equation must resolve, and every complex puzzle must at least have a messy solution. But sometimes, there isn't one. There is no hidden variable, no path forward.

"Sometimes, the universe doesn't owe us an explanation. It just is."

- Clark Kent

OBSESSION *with the* MEDIAN

when Statistical Comfort Replaces Real Competence.

Dr. Soumen Basak

In many classrooms today, especially in higher education in India, one question is heard almost immediately after every quiz, assignment, or examination: “What is the median?” Very often, this question is asked before students ask what they understood, where they made mistakes, or why they could not solve a particular problem. The median, in such situations, is no longer treated merely as a statistical quantity. It becomes a source of comfort.

There is nothing wrong with the median itself. It is a useful way of describing a set of marks. If all the scores in a class are arranged in increasing order, the median is the middle value. It tells us that half the students scored below it and half scored above it. For a teacher, this information can be useful. It gives a quick picture of how the class performed in a particular test.

The problem begins when stu-

dents start treating the median as a measure of competence. A student may perform poorly in absolute terms, but if the score is close to the class median, there is often a sense of relief. The reasoning is simple: if many others have also done badly, then my performance cannot be very bad. This is a comforting thought, but it is also a misleading one.

If the whole class performs poorly, the median will also be low. In that case, being close to the median does not mean that the student has understood the subject. It only means that the student is close to the middle of a weak performance distribution. Similarly, in a very strong class, a student below the median may still have a good understanding of the subject. The median tells us about relative position. It does not tell us about real ability.

This distinction is important because life outside the classroom

does not judge students by the class median. A job interview does not ask whether a candidate was above or below the median. A research supervisor does not ask whether a student was close to the median in an examination. What matters in such situations is whether the student can think clearly, solve unfamiliar problems, explain ideas properly, and work independently.

The danger of median-based comfort is that it can weaken the desire for honest self-improvement. Instead of asking, “Why could I not solve this problem?” a student may ask, “How far am I from the median?” Instead of examining mistakes seriously, the student may take shelter in the fact that many others made similar mistakes. Education then becomes a game of comparison rather than a process of growth.

This does not mean that the median is useless. Teachers may use it to understand the distribution of marks or to judge whether an examination was too easy or too difficult. Used properly, it is a helpful statistical tool. But it should never be mistaken for a certificate of competence.

A serious student should ask better questions after an examination. Can I solve the problem without help? Can I explain the method clearly? Can I apply the same idea to a new situation? Can I identify where my reasoning failed?

The median may tell a student that they are not alone in their weakness. But it cannot tell them that they are strong. The median is a number. Competence is an ability. Confusing the two is one of the most harmful mistakes a student can make.

Endless, Yet Still

On the benches, sat so tender,
Riddles, games, lessons, and laughter.
Drifting through language, arts, and past,
In the laws beneath, felt home at last.

On the blackboard, the world grew wide,
Met Newton, light, and chance with pride.
But at night, traveled to the stars,
Dreaming of dreams beyond the stars.

Then old enough to cross the gate,
Fought alone, defying fate.
The stars glowed dim in shadows of the mind,
The search for meaning burned, not left behind.

Greeted the giants in the timeless hall,
Yet through the empty space I fall.
But slowly, life began to light a smile,
Though it had slept a weary while.

Now seeking the law that shapes it all,
Walking the edge where wonders call.
Though papers pile and time feels still,
Holding to joy, searching still...

- Hari Govind K J (BSMS'22)

WHISPERS OF THE UNKNOWN

The glitter and the glamour of the terrestrial world, fading and swaying
Losing the last ounce of light left, they fade to the ever-expanding darkness.
The vision that came to life in this darkness, an ethereal message from the other side

There they were, in the black veil, the majestic heavens revealed in all their glory
I could spot the Hunter, the stars whose stories I grew up with, whispers from antiquity
I could imagine a gazillion more of them, whose secrets are hidden in the opaque spaces.

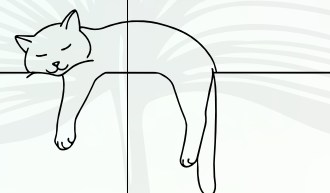
I stared at the mystical, deceiving beauty of the infinities, extending in all directions
And I couldn't help but stare with awe, gasp with wonder. Like a curious child, I asked.
Could there be another me, another life drifting elsewhere in this vastness?

Would she be dancing, feeling the sweetest tickle of an unopened present?
Would she be crying over a broken crayon or a broken heart that failed to mend?
And I wondered, would we ever meet, ever get a chance to talk, to love and to hold?
Would we ever be able to celebrate our lost pasts and shared presents? Would we?

I don't have all the answers. Maybe I will never know them.
Maybe I will wave goodbye without knowing her. Maybe that's for the better.
But now, I look at the tiny dot in the light ray, a pale blue one in the empty vastness
The home to my dreams and my stories; the rock that sees the sun and the shadow in my
heart

I wish the world saw it; We are nothing but ants clinging to a tiny, tiny rock in the sky.
Then all borders and boundaries, all the bloodshed and devastated screams evaporate
I wish we heard the whispers of the infinities. I wish we saw the miracle that we are.
As someone said, we are the universe trying to understand itself. I wish we knew it, too

- Gayatri S (MSC'23)



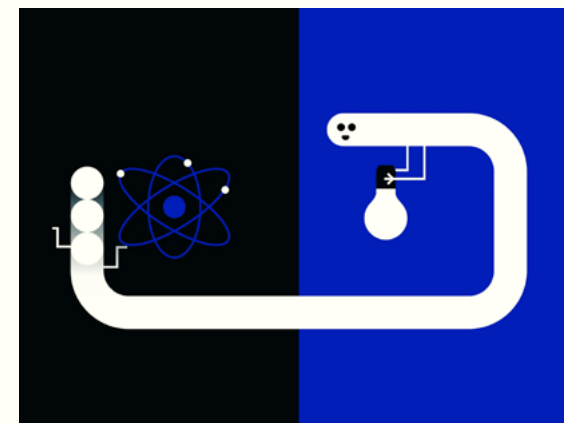


Left to right:
John Clarke, Michel H. Devoret,
John M. Martinis

Courtesy: Nobel Prize Outreach,
by Clément Morin

THE NOBEL PRIZE

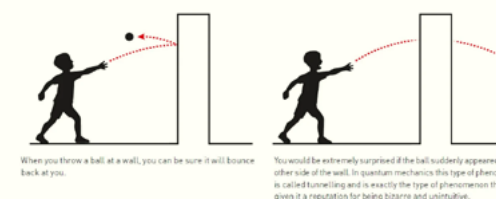
THE DISCOVERY OF MACROSCOPIC
QUANTUM MECHANICAL TUNNELLING
AND ENERGY QUANTISATION IN AN
ELECTRIC CIRCUIT



Courtesy: Johan Jarnestad, KVA Sweden

In 1963, a quiet kind of magic was brewing at Bell Labs. A deceptively simple circuit connected the macroscopic world to a quantum property long considered abstract: phase. John Clarke, Michel Devoret, and John Martinis took this idea further, shaping it into a system that behaves like a fictitious particle. This construct not only paved the way for quantum computing, but also earned them the Nobel Prize in 2025.

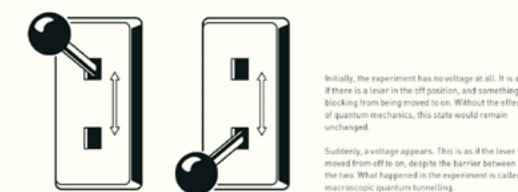
To understand the premise, let us take a brief detour into the quantum world. The space you and I inhabit can be described using three perpendicular axes built from real numbers: x , y and z . In this space, we draw arrows. We stretch them, rotate them, and combine them. Collect all such arrows, and you obtain what physicists call a vector space. Quantum systems, however, refuse to live in such modest surroundings. They demand an extra degree of freedom.



Courtesy: Johan Jarnestad, KVA Sweden

Alongside real numbers, imaginary numbers enter the picture, and the space expands into what is known as a complex vector space, or more precisely, a Hilbert space. Here, the arrows no longer represent

directions in physical space. Each vector encodes a state of the system. These vectors can be assembled to form the wavefunction of a particle, a mathematical object whose squared magnitude tells us the probability of finding the particle at a particular position, or of measuring a particular momentum. In this abstract landscape, lengths become probabilities and angles become phases. This is the phase or more technically, the phase difference that was measured in the Bell Labs.



Courtesy: Johan Jarnestad, KVA Sweden

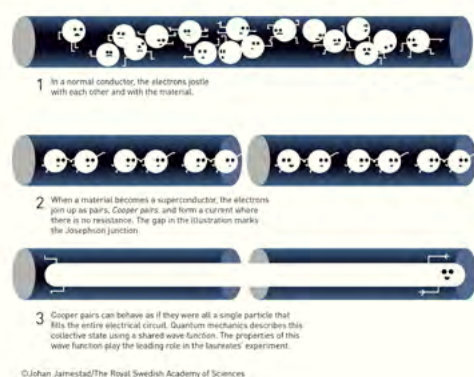
To unpack this, we begin with electrons. At sufficiently low energies, electrons, fermions with half-integer spin, bind into pairs known as Cooper pairs. These pairs behave as composite bosons, carrying integer spin. When cooled below a critical temperature, these bosons condense into a single quantum ground state called the Bose–Einstein condensate. This state is not described by many independent phases, but by a shared phase field, meaning that while individual phases may fluctuate, their relative phases are rigidly locked together. (The Josephson junction has a thin insulating barrier placed between two superconductors. Each of these superconductors exist as a Bose–Einstein condensate.)

Left undisturbed, this collective alignment remains uniform. In such a quiet state, no supercurrent flows, since current arises only from a spatial gradient in the phase. When an external bias is applied, however, the phase begins to vary across the system. This controlled misalignment manifests as a supercurrent.

The Nobel Prize, however, was not awarded for establishing that current depends on phase, but for demonstrating that quantum tunnelling governs the dynamics. Through tunnelling, the superconducting wavefunctions seep into the insulating barrier, allowing the two condensates to couple and behave as a single quantum system.



Physicists have known for almost a century that tunnelling is necessary for a particular type of nuclear decay (alpha decay). A tiny piece of the atom's nucleus breaks free and appears outside it.



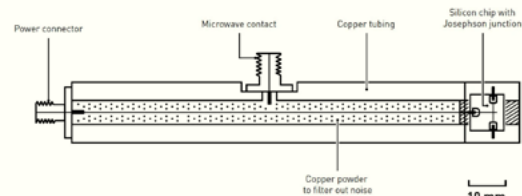
©Johan Jarnestad/The Royal Swedish Academy of Sciences

Courtesy: Johan Jarnestad, KVA Sweden

In a current-biased Josephson junction, the dynamics of the phase difference can be visualized through the tilted washboard potential. The phase behaves like a particle moving across a corrugated landscape, whose overall slope is set by the applied bias current. At low bias, the surface is only gently tilted, and the particle settles into one of the valleys. The phase remains trapped, motionless on average, and the junction stays in a zero-voltage state.

As the bias current is increased, the tilt steepens and the valleys grow shallower. Beyond a critical point, the particle can no longer remain confined and begins to slide downhill. This corresponds to a running phase and the sudden appearance of a finite voltage. Crucially, even when

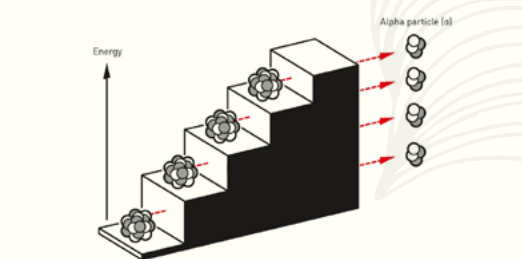
the valleys persist, a quantum phase may escape by tunnelling through the barrier, while a purely classical particle would remain trapped unless lifted out by thermal fluctuations.



John Clarke, Michel Devoret and John Martinis constructed an experiment using a superconducting electrical circuit. The chip that held this circuit was about a centimetre in size. Previously, tunnelling and energy quantisation had been studied in systems that had just a few particles; here, these phenomena appeared in a quantum mechanical system with billions of Cooper pairs that filled the entire superconductor on the chip. In this way, the experiment took quantum mechanical effects from a microscopic scale to a macroscopic one.

Courtesy: Johan Jarnestad, KVA Sweden

Experimentally, the bias current is ramped slowly from zero while the voltage across the junction is monitored. For each ramp, the junction remains in the zero-voltage state until, at a particular bias value, a sudden voltage appears. This switching point defines the escape current, marking the moment when the phase leaves its local minimum and enters a running-phase state. Repeating this procedure many times produces a statistical distribution of escape currents, reflecting the inherently stochastic nature of the escape process.



A quantum mechanical system behind a barrier can have varying amounts of energy, but it can only absorb or emit specific amounts of this energy. The system is quantised. Tunnelling occurs more easily at a higher energy level than at a lower one so, statistically, a system with more energy is held captive for less time than one with less energy.

Courtesy: Johan Jarnestad, KVA Sweden

At higher temperatures, escape is dominated by thermal activation. Thermal fluctuations assist the phase in climbing over the potential barrier, causing the average escape current to decrease as temperature rises. As the system is cooled, these fluctuations are suppressed and the average escape current increases. Below a

characteristic crossover temperature, the escape current distribution becomes independent of temperature. This temperature insensitivity signals a transition from thermal activation to macroscopic quantum tunnelling of the phase, providing unambiguous evidence of quantum behavior in a superconducting circuit.

The discrete energy levels underlying this behavior were revealed by applying a weak microwave drive while slowly ramping the bias current. When the microwave frequency matched the natural oscillation frequency of the phase within a potential well, or the spacing between quantized energy levels, the phase was resonantly excited and escaped more readily. Experimentally, this appeared as sharp enhancements in the escape probability at specific bias currents. By sweeping the microwave frequency and tracking these enhancements, successive resonant levels were resolved, exposing the quantized energy structure of the macroscopic phase itself.

The recognition that the superconducting phase is a coherent quantum variable led directly to practical devices such as the SQUID, where two Josephson junctions in a loop exploit phase interference to measure magnetic flux with extreme sensitivity. The same idea underlies superconducting qubits: an ordinary LC circuit is modified by replacing the linear inductor with a Josephson junction, introducing nonlinearity and making the circuit an anharmonic quantum oscillator. This anharmonicity separates the energy levels, allowing the lowest two states to be selectively addressed while higher levels are largely ignored. Treating these two states as a controllable quantum two-level system forms the basic building block of superconducting quantum computers.

Beyond its foundational significance, this discovery enabled numerous applications. Its impact continues to extend across modern quantum science and technology.

- Shamayela Fayaz MSC'24



INSIDE THE PHYSICS OF A PHYSICIST

INTERVIEW WITH ASSISTANT PROFESSOR, DR.

MATHEW ARUN THOMAS

Coffee, Colliders and Curiosity

What inspired you to pursue your career in physics and what continues to motivate you to push the boundaries of physics?

That's a difficult question. I don't know what motivated me to do physics, but science in general. Maybe the objectiveness in it, compared to say, from school days onwards, compared to the rest of the subjects, the objectiveness in it attracted me. It was more, different people have different callings, I guess, but that was more simpler for me. Doing science was much more simpler for me than doing anything else. But I was definitely also interested in history. At some point I also wanted to become an archaeologist, wanted to do some anthropology also. But somehow my liking or my ease of doing was better with science.

Looking back at your journey from your studies at St. Stephen's College and the University of Delhi, to your post-doctoral work, what moments stand out as a pivotal in shaping your career?

St. Stephen's College had its involvement in it. There was an annual meeting every year, where a lot of people used to be invited individually. Talks were given by a few eminent people invited from all over the country. It was called Popli Memorial Lectures. And many people from various institutes had come and gave lectures there. And that was also an opportunity to meet likeminded people from the entire university. So many famous people would come. In St. Stephen's College, we also had something called a problem-solving club, which was very interesting. So people would take problems, this was bachelors, it was a BSc course and then it was a full-year course. So these were completely run by seniors, some third years or second years would run this particular club.

They would take problems from wide range, or you make your own problems and you come there and solve it. For example, if we were doing Newtonian mechanics the problem would not be solved using Newtonian mechanics, but you can involve say Lagrangians or Hamiltonians. So at that time when it was first year, it was completely new. But you learn techniques, you learn some physics in it, and that was quite good. Not to say my faculties from St. Stephen's College had their own large involvement. Professor, if I recall the names, it's Professor Bikram Phookan, and Dr. Abhinav Gupta. They had a major role in shaping the area. I remember, Prof. Phookan almost took two months of Special Theory of Relativity in first year. It's a year course, so that's one full semester of Special Theory of Relativity, which covered quite a bit and quite in depth. That was wonderful. And even from the mathematics department, we had a first year of real analysis, second year of real analysis course, which were pure mathematics course and they were very deep and quite interesting. The faculties, of course, they were good. They had their own research backgrounds. The University at that time, used to be quite good, very good, research wise, and also quite involved. Like T.R. Seshadri, my guide was Debajyoti Choudhury and Daksha Lohia. They were quite a lot of people whom I could interact with. There was a vibrant research community there. It was quite vibrant compared to rest of the universities in the country. And that's probably the pivotal point, the formative years.

Did you always know you wanted to do physics, or where you interested in other disciplines like chemistry or math?

Chemistry, I started liking organic chemistry when I really started preparing for my exams. And inorganic chemistry was very difficult for me. I just could not

understand inorganic chemistry. But organic chemistry, I started liking when they started talking about electron transports. So between two

bonds, how the electron transports to other. But more than chemistry, I was interested in Biology. For example, In gene dynamics of the genetic system, how the T and T structures come together. So the point is, my liking is not towards any subject in that matter, but more objective or logical things. So if you have to by-heart a lot of things, somehow my brain would cut off. So the aspects in biology, chemistry or maths and physics that I started liking was more objective things. Something we can logically connect. So I don't want to say whether I did not have a liking towards any other field. But yes, to a particular part of that field that requires that kind of analysis, I always liked. Somehow physics had it all. Physics combines it all. You don't have to by-heart a lot of things. But chemistry on the other hand, if you want to do a proper thing, you have to by-heart a lot of things. Even biology, you have to remember a lot of things. Some of physics, you just didn't need any remembering stuff. So overall, when you analyse things, physics is there. That's why I liked anthropology for a bit, or sociology. They are also sciences, so they also have a structure to it. There is a way of analysing things. So I had a liking towards them.

Could you elaborate on the core themes of your current research at IISER TVM? Particularly in high energy physics and particle phenomenology.

Initially I started with extra dimensions. Warped extra dimensions and some more formalism kind of things. How will you construct a six dimensional geometry? Which is warped or flat, doesn't matter, but there are certain conditions. If you have six dimensions, for sure we are not living in six dimensions. We are living in four dimensions. How do we know? - because we know the 1 by r potential. Electromagnetic radiations are 1 by r. If you go in different dimensions, that potential is something else. 1 by r squared, 1 by r cubed. If you go to lower dimensions, it will be log of r or a constant. So how the potential goes is crucial. It

depends on which dimensions you are in. And just by looking at the sun and how much time the light takes to come to you, you can actually say it's 1 by r potential. And that's a verification of it. So even if you suggest that we are living in some extra dimensions, it has to somehow come down to four dimensions. Somewhere the symmetry of six dimensions has been broken down to four dimensions. So how do you say that? What symmetry and what kind of repercussions do you have when you have the six-dimensional Lorentz symmetries being broken down to four dimensions? That's something that I was studying initially. That was during my PhD time. And moreover, if it's a geometry, it has to be stabilised. Otherwise at some point it will collapse. So how do you stabilise this kind of geometry? After that, how do you explain fields? As we know, the gauge field, fermion field, etc. How do you explain those fields in this? And how do I relate it with the current experiments? So I had a part in my career as one is a formalism, and the other part is how to connect these formal stuff with the current experiments. Say, at Large Hadron Collider, if we see something or we don't see something, how does it relate to this kind of formalism?

Later on, after joining IISER TVM, I started working on phenomenology. Initially it was more on flavour physics. Flavour physics means different generations of quarks or leptons. And sometimes, various beyond standard model particle physics, do propose some sort of flavour violations. Some muon going to electron. Say, muon going to 3 electrons, something like that. Or, some will enhance some amplitudes that we have seen, like Kaon K0 bar oscillations. In a warped extra dimensional setup, how do we limit these kind of models using these indirect experiments? That's something initially I had worked on. Then in 2021, there was a result by Super-Kamiokande that they've almost seen 11 events of neutron anti-neutron oscillation. Neutron anti-neutron oscillation is a violation of baryon number by 2 units. Baryon number violation is crucial for our existence. So if you want to see a matter-antimatter asymmetry, one of the conditions to be satisfied is this baryon number violation. Unfortunately, we have not seen any. Why? Because proton and neutron

are the most stable objects. Neutron do decay to a proton. Beta decay exists. But if you have a proton, just a proton, it will never decay. But even a neutron decay is not a baryon number violation. Neutron goes to proton. So baryon number is constant. But a baryon number violating of a neutron, decay of a neutron, is also not observed. These are observed, meaning that the timescale is 10 to the power 32 years. That's huge. The current limits are huge. Which means, if you look for a neutron and a proton, they will never decay. And if they do not decay, then the question is how are we created? How are we existing? If the matter and antimatter do not have asymmetry, we should just annihilate. We are all just radiations. We don't have our own existence. So 11 events, it is not a significant amount of events, because the background is also some 9.2 or something. So 11 is just above the background of neutron antineutron oscillation, which is very interesting. Based on this, a lot of other experiments were also proposed. There are new experiments coming up called high beam, NN bar, which particularly, very specifically, look for neutron-antineutron oscillation, or things called mirror neutron oscillation. Let's say if I have a dark sector, which everyone knows. There is a dark matter, we just do not know what it is. And if there is a neutron sitting there in the dark sector, whether our neutron can convert to that, so, such kind of research. Since that experiment was crucial, and a lot of experiments are being proposed, and it's a very vibrant field right now. Seeing that particular result, I kind of shifted to that. And now my current research has different facets. I also work a bit on ADSQCD. It's not the top-down string theoretic method, but bottom-up holographic method, which is field theoretic. You go to five dimensions and see what QCD can you get from there. So that's also part of mine. And a bit of, recently, in the last three, four years, there has also been some push towards machine learning in ADSQCD, so I'm doing that.

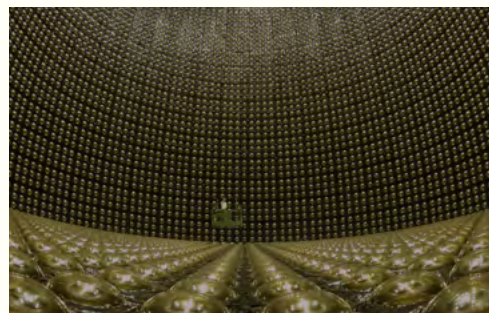
Can you explain why broken symmetries are of importance?

We all are. Just look at the presence of a table. It breaks the symmetry, right? The translation symmetry is just broken. I cannot go beyond it. It's like we are living in a broken world. We don't have perfect symmetries. The only symmetry you can say in the universe, even, is in parsec scales. In 100 megaparsec scales, it's homogeneous. But if you look at locally, the isotropy is lost, everything is lost. But these are just fine. It's not breaking anything. But beyond that, in particle physics, our understanding of particle physics, it comes from experience. It's a phenomenological topic, right? We don't have a topdown model. We have a bottom-up approach. So in a bottom-up approach, we live with experiments, and we try to explain things that are related to experiments. So in the history of particle physics, if you look at that, at some point, people realise there's a weak interaction. There has to be something called a weak interaction. That violates C and P, but conserves C. And say, for example, muon going to electron, a muon neutrino and an anti-electron neutrino. That was observed, and beta decay was observed. And this thing said that there has to be something called a charge current. It cannot be a neutral current. It has to be a charge current interaction. Now you write your theory based on this charge current interaction. So happens the only thing you can write is a SU2. You cannot write a U1, because U1 will be a neutral current interaction. This will be a SU2, a charge current interaction. Now you have a charge current. But unfortunately, in none of my experiments, I ever see a charge. If you have charge current interaction, then whatever mediates them are SU2 bosons. Let's say, some SU2 bosons are there. But nowhere in my experiment, I see a massless SU2 boson. But I know, light, for example. How do I know that it's massless? It's infinite range. If it's infinite range, it has to be massless. So then the question asks that what happened to this. I know there's SU2, there has to be SU2, because I see weak interactions very dominantly, and

I also see a massless U1 interaction. But how do you connect these two? So then people proposed. So that's it. Over the years, people proposed various things. And finally, Jojoi, Salam Abdusalam, and Wilczek, etc., they proposed this SU2 cross U1 theory. Peter Higgs, with others, finally said that it has to be broken down to a U1 electromagnetism. So, somewhere this breaking becomes very important. Even simpler, for example, that we all know, ferromagnetism. You know there are ferromagnetic objects, right? There's no doubt about it. How do you explain their existence? You know, you keep it in a magnetic field, you have a certain magnetic field, even if you switch off the background magnetic field. How do you explain it? That is explained by a symmetry breaking of a Z2. A parity symmetry being broken spontaneously. That creates domain volts inside, and that localises a spinor in a particular direction, and that's why you have a certain magnetic field. So somehow we are living in a world. We have masses, right? Not our mass, our mass does not come from the minute electrons mass, or anything brought on. But the mass of fundamental particles, we know they are massive, right? X of a neutron. Now we know they are massive. So, there's no other way to dynamically explain their masses. Say, y is electron's mass, 0.5 MeV, and not keep on increasing. How to control electrons. So the only way in which we know how to give it is through something called a spontaneous symmetry. So that's where the breaking is very important. In our universe, we have all broken systems, most of them.

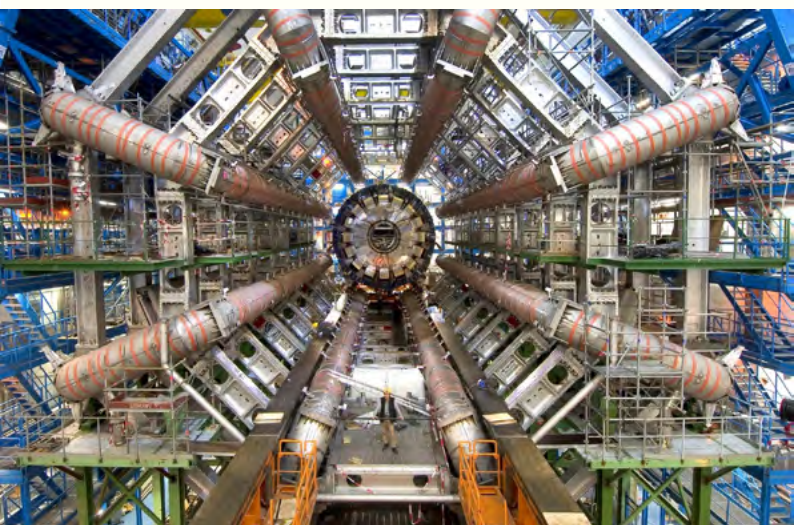
How do you bridge theoretical models with experimental data, and what challenges do you face in ensuring both approaches inform each other effectively?

Okay, so that's my career. So, whatever you make theoretically, it has to be seen. There's no other point of making things theoretical models if it's not being seen. So, for example, Karl Popper, in his autobiography, he even mentions that why he liked science more than any other dogmatic things, like communism, for example. It's because the rest are dogmatic. He listened to a talk by Einstein where he said, if you see a redshift, then that makes my theory part of a bigger theory. But if you do not see a redshift, then my theory is false. So



Super-Kamiokande,
Courtesy of ICRR, Univ. of Tokyo

Karl Popper goes and explains that at best, the scientific theories are limitations of some bigger theories, but one experiment can falsify it all. So that's the connection between science and experiment, because the science works. As a scientist, what you do is you're trying to falsify a model. You're never trying to prove a model. You're trying to falsify a model and see whether that model fits in as a limitation of a more beautiful model. And when I say beautiful, it means something like a Occam's razor. The least number of parameters explaining the most amount of things that you see. But that explanation, or that kind of thing that comes from experience, you match it with the model and see whether that is correct. Finally, it all matches. It all comes down, boils down to experience. And that's, in a very short way of saying what I do. But most of the part, I'm more in the formal sector, but of course, they are all connected.



Large Hadron Collider
Courtesy of SLAC, CERN

What are the most pressing unanswered questions in particle physics that your research, or research throughout the world is aiming to answer right now.

So, my particular question, right now I'm more interested in low energy experiments. The two most pressing things, in my opinion, is dark matter and beta number evolution. We know they exist cosmologically, because of course beta number evolution, we all exist. And dark matter, again, we all exist. Because without dark matter creating a potential, various large structures wouldn't have been formed,

etc. We know it from cosmological simulations and all these things. We know it. So, they have to be there. We don't know a particle origin for this dark matter. How are they? What are they? That's something we don't know. So, these two, somehow, and in my opinion, they're somehow talking to each other. They are in cohorts. They are playing a game with us, I guess. But somehow, if you try to manage them to come together, some dark matter induced baryon number violation, dark matter carrying an anti-baryon number, which takes away some of the baryon charges to dark sector. There are a few proposed models, quite interesting models. So, in my opinion, those are very important things. Similarly, the world is also looking for something called axions. Axions are pseudo-Goldstone bosons. They're pseudo-scalars. The importance of axiom is that, as we just talked, there are always broken symmetries. We always have broken symmetries. And pseudo-

Goldstone bosons are the remnants of those broken symmetries. And, of course, in our universe, we say, Standard Model of Particle Physics is a recipe book. It's a wonderful recipe book. You have not seen anything beyond it. But it is not a beautiful theory. It has huge number of parameters. So, you expect

some bigger symmetry to be there, which broke down to this particular symmetry. So, then, Goldstone bosons play an important role. Based on that, axions. Axions are nothing but Goldstone bosons of some symmetry breaking, and all over the world, there are experiments being proposed, new experiments coming up to detect axions coming from Sun, coming from galactic centres, from various places, etc. In the Large Hadron Collider, after seeing Higgs, we have not seen anything conclusive. There are always some resonances that you see, but statistics. When you use statistics, when you do an old data compilation, the significance of all those resonances

comes down. That's what being seen. So, I strongly believe that low energy sector holds a key to understanding the new physics.

Could you share insights on any recent research or breakthroughs from your research that you believe are especially significant for the field of beyond standard model physics?

As I said, the superconducting current is the result of 11 events, seeing 11 events are crucial. Now, the new experiment will come up for Hyper-Kamiokande, which will go much beyond the sensitivities of super-kamiokande to see whether they are right. And India is also planning to come up with some experiment in a bit long term. Let's see whether they also look for baryon number violation, but they are looking more towards neutrino physics. Currently, this, I believe, is the most significant result in my mind.

How is AI/ML being used to prove or disprove results in these experiments?

So currently, LHC, for example, is machine learning. I only remember some BDT analysis, but people have gone beyond it. It's a black box, so you do not know exactly what happens. It depends on who trains or what trains. It becomes model dependent, but people use it. So if the whole community is using it, then there has to be some reason for it, right? Is that I do not know. I have not started doing it or I'm not an expert. I don't know what it is, but what they mostly do is statistics. So it is, the question is to how well can you differentiate the signal from the background? So they do a statistical analysis. That is not to try to explain a model or something. It's not a theoretical analysis. Statistical analysis and machine learning is done for that. You train the data with some statistics and you explain that. On the other hand, the kind of machine learning that I am right now studying or doing with my PhD is more on solving some differential equations. We have data that are masses of pions, masses of raw mesons, different resonances of raw mesons, different resonances of different mesons, A1 meson, F2 meson. So we have a lot of tensor mesons, all those resonances. Some of them we have seen, some of them are predicted.

What we do not know, the kind of the more good theory right now is of QCD, low energy QCD is something called AdS QCD. You have AdS in the bulk, it's five dimensional theory, AdS in the bulk and on the brane, on the surface, you have a QCD, a low energy QCD. So the mesons are only living on the surface and correspondingly, some gauge fields which are perturbative are living in the bulk. So some easier calculation in the bulk can lead to the understanding. Mass spectrum, for example, in the bulk of these gauge fields will correspond to the raw meson masses on the surface, on the CFT or broken CFT, QCD. So that is the analysis. But what we do not know in this, from a bottom up approach is the background gravity. We have no understanding of what this AdS is, right? So you propose something. So people have proposed something based on, some geometries based on older models, like from Randall-Sundrum they have proposed some five dimensional warp geometry. So based on those kinds of things, people have gone ahead with it. It matches okay with the resonances, but not with order. So the statement is the following, that as heavier resonances go, they are all linear. So then you try to match, you try to find a geometry or some fields, scalar fields, etc, some breaking etc., such that you match these things. But the question that we are asking, can you train a model, train an AI to predict the masses? Essentially boiling down to solving a differential equation by training the data from the masses of say, rho meson, or other mesons, etc. And thus can you explain, can you come up with the background gravity, or something called the electron field, can you see how it looks like? So that is a different kind of use of AI, or I will not say AI, machine learning, ML in physics.

Can you describe your approach to mentoring students and young researchers, and how does that environment foster innovative ideas? Could you share how your mentors, Professor Debajyoti Choudhury and Professor Sudhir K. Vempati, influenced your research and profession?

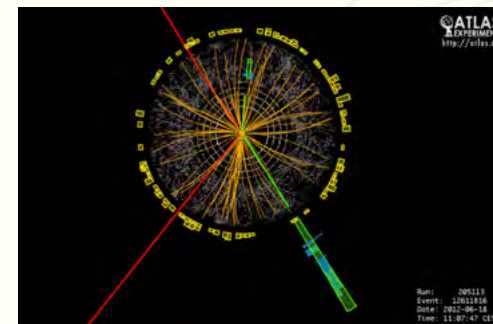
Debajyoti and Sudhir, very unique people. Debajyoti, I went to him, I discussed with him, and he's more a particle phenomenologist, but he also

has his interests are everywhere. Right now, I think he's doing magnetic, you know, dynamos in cosmology. He is now going away from particle physics to cosmology and magnetic fields in cosmology, the seed magnetic fields in cosmology. Last time I heard his talk, he was talking about that. He was talking about magnetic fields in cosmology. So, I went to him, I said, I'm interested in working with you, and first told him I want to work on quantum gravity. You know, young guy, full of excitement, string theory, interesting, highly formal stuff. He was one of them. He looked at me and he said, okay, I'll ask you to read up some, something else, see whether you are interested in that. And I remember, he gave me, he asked me to, that time, extra dimension was a hot topic, you know, understand beyond standard model through extra dimension or supersymmetry was a very hot topic. I told him I'm more, you know, I have better analysis in visual looking, understanding things visually than writing things down. So, he said, okay, you better work on extra dimension, not supersymmetry. But which is not true, I mean, supersymmetry also is a very beautiful, it's a very visual subject, nevertheless. So, the beginning, he asked me to write a Lagrangian. He's more of a phenomenologist, right? So, he asked me to write a Lagrangian, an extra dimensional setup, with fields localised on the brane. And he went, he's a busy man, he goes for his meetings, etc. So, he went for his meeting. And so, I sat there. Five minutes later, I wrote the Lagrangian. How difficult can it be, right? A four dimensional Lagrangian, put a delta function next to it. Some four dimensional Lagrangian, and then delta of xy. It's a Lagrangian localised at xy equal to zero. And then you put a delta plus a same Lagrangian, delta of xy minus pi, or pi r. Now you have a localised at the pi r. Then I went around looking for him. I could not find him, he was busy in meetings, lot of meetings. So, I came back and I started thinking, is this all that he asked me to do? This is very simple. Put a delta function next to a particular four

dimensional Lagrangian, it becomes a five dimensional Lagrangian, but localised on the brane. Is it all? Then I started reading on it. I started now reading into very old papers, which exactly talked about the Ising model in five dimensions. It's not Ising model, it's a Z2 symmetry breaking, but instead of spins being localised, now fermions are localised on a brane. It's a very dynamical way of localising fermions. Not like what I did at delta function next time. So, this is highly dynamical and I mean, you get fermions, you localise left-handed fermions on one side, right-handed fermions on the other side. It's beautiful. Along with it, I started reading on something called Dvali-Shifman mechanism which is about localising gauge fields. I started

thinking, is this what he wanted me to do? The other thing was five minute job. So, I think five days back, he came back and then I met him. I showed him this, you know, yeah, I'm reading this and these are the solutions. And

he looked at it and said, yes, I just asked you to put a delta function next to it. So, then he said, there is something quite turning my life, you know, he said, okay, now I think you're not a phenomenologist. He said, you're more a formal guy, because my liking is more towards understanding what are the dynamical mechanism that causes this, rather than using this Lagrangian to see what are the effects in LHC. So, he wanted me to do that, you know, take this Lagrangian and see what are the effects in LHC or some other colliders or some, which is more his favourite topics. But rather I went in a completely different direction to understand what are the dynamical ways of localising and how does a breaking happen, how does a Lorentz symmetry happen in from five dimension to four dimension. This is a different set of topics, more formal topics. So that's what it started with. So, Debajyoti gave me my freedom. He told me, okay, now it seems like you are a more formal person, you do formal stuff. And that's what we worked on since. And so, he gave me my freedom, right? He didn't tell me



ATLAS Experiment
Courtesy of CERN

that, no, no, no, this is not what I probably could have done, right? No, he would have said, okay, no, this is not what I meant. You work on this, right? Write a Lagrangian, write a delta function, look at the collisions in a LHC or something like that. He could have said that. And I would have hated, maybe I wouldn't have liked doing it, but I could have, I would have still done it probably, but he gave me his freedom and that's my start of my intellectual freedom, right? I was not forced to do something, I was doing something I like. So, that's my experience with Debajyoti. And similar things with Sudhir. Sudhir was also more interested in me talking to them than he telling me. He is a very informative person. He works on flavour physics and all the conversations with him, the insights that he has had or he has are just beautiful. How you approach a particular subject in flavour physics is something new to me. I had not worked on flavour physics before, never worked on. And when I went to IISc, it was purely flavour physics. Sudhir's insights were beautiful and I learned a lot from him. He wouldn't dictate something I would do, I need to do. He would rather say, okay, I have something very interesting, go read it up and tell me. And so I would read it up, go back to him and say, okay, this is something that seems interesting. Can you work on it? And he said, okay, let's work on it. I hope I'm trying to do that with my students also. If they're interested in something, go explore it, come back and teach me. That's kind of my attitude.

Balancing rigorous research with teaching and administrative responsibilities can be challenging. How do you manage these demands and what advice would you give to young academics?

I don't have that many administrative duties. Others around the department or, you know, they have much, much more. I'm mostly here. Teaching is fun. And we have only three hours of teaching, right? A week, we have only



Construction of Hyper-Kamiokande
Courtesy: ICRR, The Univ. of Tokyo

three hours. So that's ideal, I believe. So faculty teaching three hours, you need some time for preparation and then you teach. And then you come back and you do your research. So this is a very free environment. We get more time in research. It's wonderful.

As you look ahead, what are your aspirations for both your research and your contribution to the broader scientific community? What do you think will be your major contributions in the upcoming year in the scientific community?

So, my research topics are definitely there. But now, see, unlike my PhD or postdoc time, you try to make a theory more beautiful, and then you model very nice models, and then you try to match it to the experiments. Now I'm more on a lookout for experiments. And not just in particle physics, say gravitational waves or something. Gravitational waves have its own importance in particle physics. And so now I think, somehow my brain has switched gears, and now it's looking for more, to see something, whether there are interesting experiments coming up in physics, in this thing. Somehow, at some point I was even looking at, say, condensed matter. Not even condensed matter, you know, material science. Not even the theoretical condensed matter, which is very close to field theory, which is field theory. But

something like, say, material science, whether few of the materials that we have, how good, how well they can, again, in baryon number violation background, how well they can detect some baryon number violating currents or something. I was looking at some C686. I've never done this kind of thing before. I was reading a paper on that. So somehow my brain has switched gears now to something more experimental, looking at experiments, and more phenomenology stuff. I'm not an expert on that, but somehow I'm very interested in that. I've also started liking towards more formal biology. Very interesting things in biology are there. How to express the entire neural circuits as electrical circuits and things like that. Finally, the problem is that you cannot write a differential equation for a biological system. It's too complex to write a differential equation. Extremely non-linear. We can solve linear equations, but non-linear equations are still difficult. So you use perturbation theory to solve non-linear equations. The good thing about an electrical circuit is that you can always convert it to a differential equation. So you can look at a certain controlled way of looking at neurones and some activities in biological systems with this kind of thing. So I'm quite interested in that.

crystallizing LIGHT

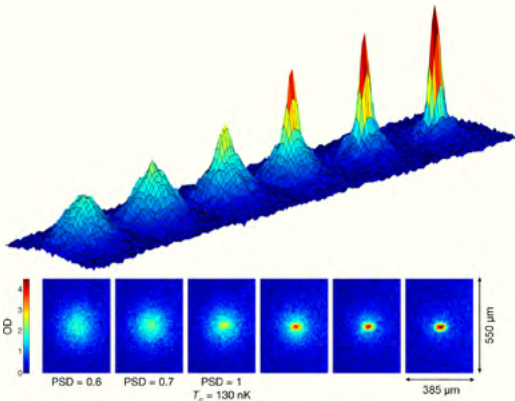
Niranjan Jayprakash (BSMS'21)

Last summer, when Manuel Landini visited Antonio Gianfate and Trypogeorgos in Italy, little did they know that the friendly discussion about Supersolids and Optical Parametric Oscillators would create the first photonic supersolid at room temperature. This nature paper published just 3 months back can be viewed as a landmark discovery in our effort to understand the emergent properties of this exotic state of matter. As a fourth-year physics graduate who was interested in polaritonic physics, I was discussing the creation of the first photonic supersolid at room temperature, which was earlier realized only in ultra-cold atoms, to a curious junior, who despite being a physics major, instantly replied, "Chi Bhai !!, Dumb it down for me !". So, in a humble attempt to explain this incredible discovery, akin to the blind leading in the dark, I will try to explain what actually went down and why this is a big deal!

We are familiar with the three states of matter from everyday life, solid, liquid, and gas. But exotic states form as temperatures hit near zero kelvin, revealing matter's quantum mechanical nature. The supersolid phase is a state of matter that combines the periodically ordered nature of a solid

lattice and can flow without friction like a superfluid. Imagine a river that flows smoothly while maintaining a perfect ripple pattern, which is supersolid for you. The similarities between a parametric oscillator and a supersolid prompted researchers to investigate the possibility of a photonic supersolid.

To better understand the experiment, let's have an analogy. Imagine you are in a crowded theatre where three seats are empty in the first row: one in the center and 2 each on the sides, while all the other seats are filled. Naturally, all incoming audience members will try sitting in the middle seat as it offers the best view.



Bose-Einstein Condensate Graph
Courtesy: Jet Propulsion Lab, NASA

Now, we equate our quantum system to a theatre and bosonic particles

(photons) as the audience. In that case, we can view the middle seat as the ideal state the photon would want to occupy, or as a physicist would call it, $k=0$. The seats represent quantum states available to photons, while the audience members are the photons themselves.

Bosons are quite communist in nature as, unlike their cousins called fermions, they have no reluctance to share the same state, which means more than one member of the audience, aka photons, can occupy a single state. So when the researchers increased the laser power, which meant more photons were entering the system, they reordered themselves to form a coherent phase of light with a

fixed lattice pattern of density, forming a supersolid. In reality, the two states on the side are of equal energy formed by scattering photons in pairs of equal energy to nearby states. This created a system whose spatial density of photons is characteristic of a supersolid.

This discovery has helped us understand supersolid as an isoenergetic parametric scattering of particles, in simple terms, scattering of particles of equal energies to adjacent states. This is observed in almost all non-linear photonic systems. This experiment will help us further probe and develop our understanding of this novel state.

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Cosmic Cliffs of Carina
This infrared landscape reveals NGC 3324, a stellar nursery previously hidden by cosmic dust. Captured by the James Webb Space Telescope, these "mountains" - some 7 light-years high, showcase emerging stars and ionized gas being sculpted by intense ultraviolet radiation.
Courtesy: NASA

STARS AND BEYOND

Nilesh Pandey (BSMS'20)

Why are stars several light years away important?

Have you ever wondered where the oxygen, carbon, and nitrogen that make up over 96% of your body come from? Although stars burn far away, they're the source of these elements. Inside early massive stars, nuclear reactions forged them, only to be expelled into space through the most enigmatic explosions in the universe-supernovae. This is why we often say we are made of stardust!

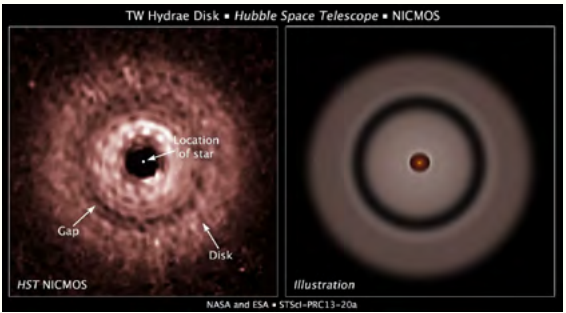
It is hypothesized that any element in our bodies heavier than iron has passed through at least one supernova (Quite cool ;)).

Detection of technetium (element 43 in the periodic table) in red giants confirms that stellar nucleosynthesis continues even today. But where can we find this stardust? Scientists have uncovered it in exceptionally pristine, primitive meteorites that escaped alteration on asteroids during the early Solar System. But this article is not

about nucleosynthesis or supernovae; rather, it is about how stars and planetary systems are born in stellar nurseries, building the foundation of life as we know it. My master's research at the Indian Institute of Astrophysics provided deeper insights into the process of star formation and its related stellar populations in our Milky Way galaxy.

Stars and Planet Formation

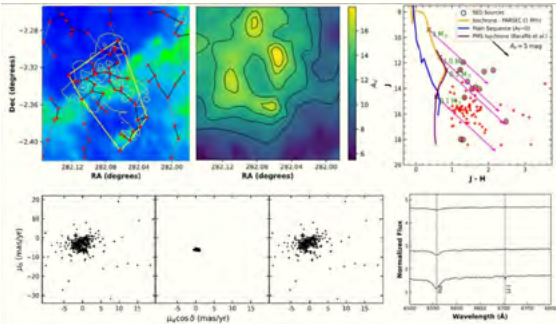
These giant, fire-breathing nuclear engines are born in stellar nurseries called nebulae, and their formation relies on two main ingredients - gravity and cold dust. Clumps of cold dust residing in the molecular cloud reach a critical mass (Jeans mass), allowing them to collapse under their own gravity. More precisely, a gas cloud's self-gravity can cause perturbations on sufficiently large scales to become unstable. The collapsing



Protoplanetary disk of dust and gas whirling around the nearby red dwarf star TW Hydrae.
Courtesy: HST

cloud fragments into lower and lower mass pieces, each collapsing on its own free-fall time scale. The process ends when the fragments become so dense that they are optically thick to the radiation cooling them, so the

radiation can no longer escape, and the temperature begins to rise, stabilizing the collapse and forming a protostar.



A technical overview of the star-forming region, combining color-magnitude diagrams and extinction maps (top) with proper motion and spectral analysis (bottom) to identify and characterize young stellar populations.

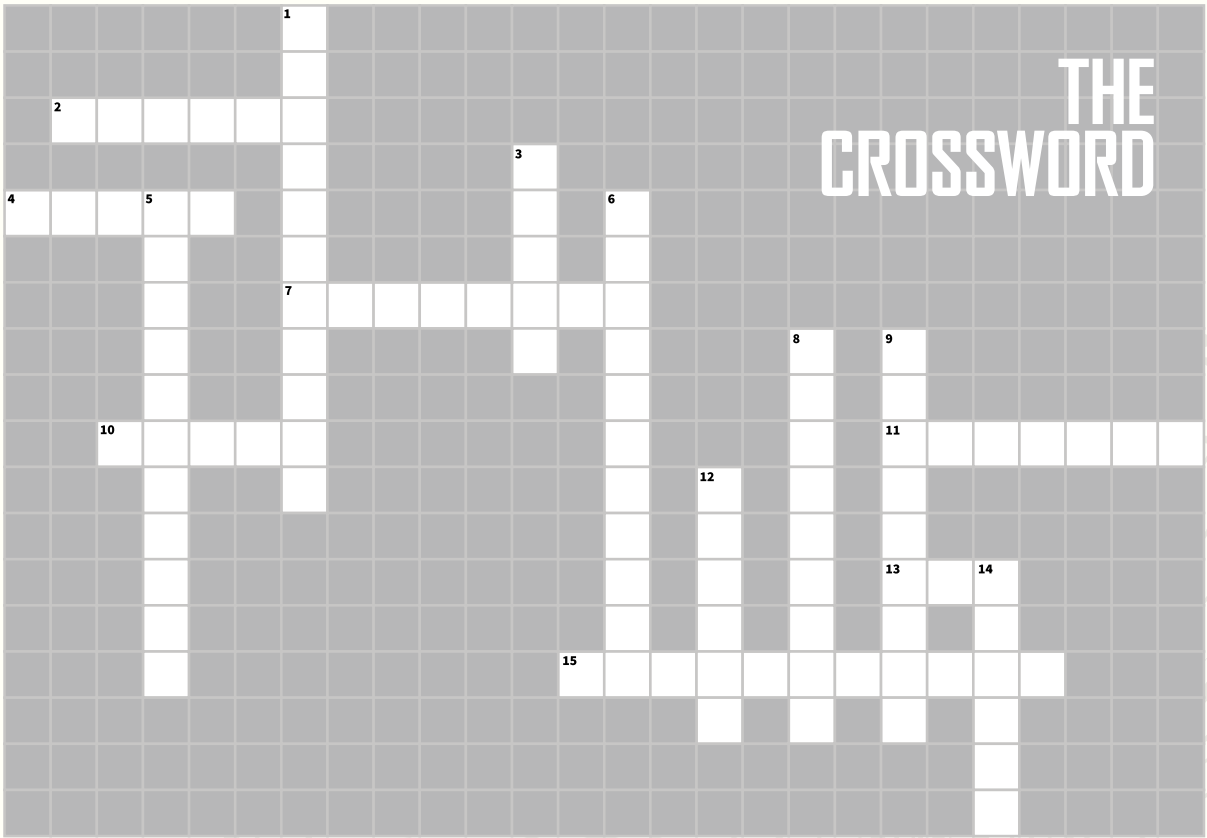
During collapse, conservation of angular momentum causes in-falling gas and dust to spread out into a rotating protoplanetary disk around the young protostar. This disk feeds additional material onto the star and acts as the nursery for future planets. Small dust grains collide and stick together, growing into pebbles, then planetesimals, and eventually into full-fledged planets through a combination of coagulation and gravitational attraction. Interestingly, we can resolve & observe these disks in detail with interferometers and telescopes such as ALMA and VLT.

Gaps and my current research

Back in 2001, a brief flare near IRAS 18456-0223 revealed only a faint nebula and a few dim stars. In my research, I've re-examined this region using Gaia DR3 (now at just 606 pc

away) alongside infrared data from 2MASS, UKIDSS, Spitzer, WISE, Herschel, and our own spectra. We found 89 young stars - nine still in their protostellar cocoons and eighty with dusty disks - grouped into tight clumps about 0.5pc across.

What really stands out are the clear gaps cutting through the cold (10–13 K) filaments: empty lanes where gravity and young stars have cleared out gas and dust. These gaps show how pockets of emptiness shape where and how new stars form, offering a simple but powerful view of cluster evolution. The study is finalized and will soon be sent off for journal publication.



Across

- 2. Sun's energy source
- 4. 1933, Nobel Prize for Physics
- 7. Atomic scale unit named after a Swedish physicist
- 10. 105 in Willow
- 11. Sudden urge or push
- 13. Volt squared per watt
- 15. Doubt Heisenberg's principle

Down

- 1. Where physics offers no answers
- 3. As small as you can get
- 5. Reduction in strength
- 6. Total energy operator
- 8. Runniness
- 9. Physicist associated with light scattering
- 12. In succession
- 14. A glitch in it can cause déjà vu

ENCODING Life

Information Transfer in Gene Expression using Information Theory

Akshat Kerni (BSMS'23) & Sujith S (Shiv Nadar IOE)

1 Introduction

Gene Expression can be understood in terms of Information Transfer using Information Theory. For this purpose, a simplified model of Gene Expression has been considered [1]. This includes an **input concentration (c) of Transcription Factors (TFs)**, which can bind and act as activators (or repressors) of gene expression and **the resulting output expression level (g)**.

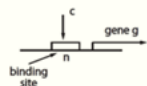


Figure 1: Gene Expression Model

Gene expression is a stochastic process, resulting from various sources of noise like transcriptional and translational bursts, input diffusion, and output noise. Therefore, the steady state input-output relation (between c and g) cannot be realistically modeled in a one-to-one deterministic model. Given an input c , **the output g will on average take the value $\bar{g}(c)$ but dynamically fluctuates around this value**. In this framework of probabilistic distributions of input and output of gene expression, **the information transferred can be quantified using Mutual Information** to optimise the regulatory network. Optimizing the network in this context refers to finding the input distribution $P(c)$ which would result in the maximization of Mutual Information.

2 Theory

Mutual Information: It quantifies how much knowing one variable reduces the uncertainty about the other. (can be expressed as a difference in Shannon entropy). It is a unique assumption-free measure of interdependency between c & g . It obeys data processing inequality, is symmetric, always non-negative & measured in bits or nats.

$$I_m(c;g) = \int dc P(c) (S[P(g)] - S[P(g|c)])$$

Here, $S[P(g)]$ is the entropy (uncertainty) of g , and $S[P(g|c)]$ is the average entropy of g given c . We can also define Mutual Information as Kullback-Leibler Divergence between Joint distribution and product of Marginal distributions. KL Divergence quantifies Information Loss from Input to Output.

$$I_m(c;g) = D_{KL}(p(c,g) \| p(c)p(g))$$

Channel Capacity: Maximum value of mutual information transferred in a noisy channel.

Blahut-Arimoto Algorithm[2]: Iterative numerical method to compute the channel capacity and the corresponding optimal input distribution $P(c)$.

Fisher Information is a core concept from estimation theory and is typically interpreted as a measure of the information provided by an observable about the corresponding unknown stimulus (parameter). It is defined by the Cramér-Rao bound:

$$\text{Var}(\hat{c}) \geq \frac{1}{I_f(c)}$$

where $\hat{c}$ is the estimator of c . Fisher Information is defined as

$$I_f = \int dg \left(\frac{\partial \ln p(g|c)}{\partial c} \right)^2 p(g|c)$$

Though called "information," Fisher information is not measured in bits. Instead, it has the units of the inverse square of the parameter being estimated. In this context, since c represents concentration, $I(c)$ has units $[c]^{-2}$. Apart from making use of the Cramer-Rao inequality, Fisher Information can be interpreted as the sensitivity of the output to a change in input at that concentration based on the input-output curve.

Jeffreys Prior: The analytical optimal input probability distribution $P(c)$ (maximizes $I(c;g)$) for a single parameter system. Used to verify the numerical result obtained from BA algorithm.

$$P_J(c) \propto \sqrt{I_f(c)}$$

Intuitively, Jeffreys Prior can be understood as, allocating higher probability to input values where the system is more responsive or informative based on the sensitivity interpretation.

Hill function: Non-linear relation $\bar{g}(c)$ used to model gene expression.

$$\bar{g}(c) = \frac{c^n}{c^n + K_d^n}$$

K_d - Dissociation Constant and n - cooperativity.

The system considered in our project: $g(c) = \bar{g}(c) + \eta$, where η represents an additive Gaussian noise around the mean $\bar{g}(c)$ as shown below. We applied BA algorithm to find the optimal $P(c)$ and verified the result with Jeffreys Prior.

3 Results

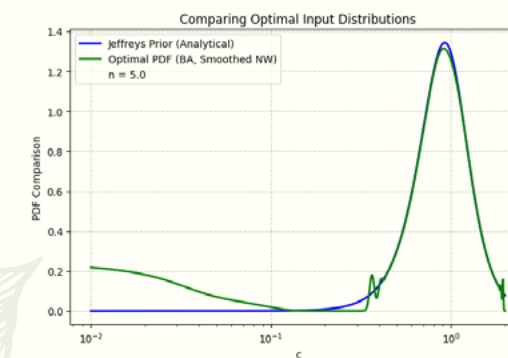


Figure 2: For constant noise $\sigma = 0.002$, Capacity: 6.873 bits.

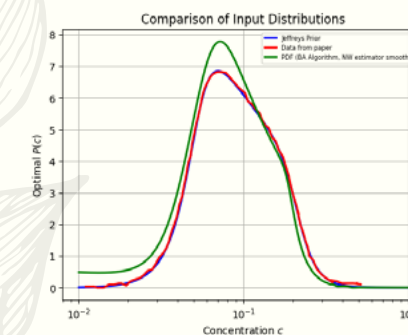


Figure 3: Optimal $P(c)$ comparison for noise model: $\sigma_T(x, \bar{y}, \bar{y}') = \sqrt{\alpha_1 x^2 + \alpha_2 \bar{y}(1 - \bar{y}) + \alpha_3 \bar{y}' + \alpha_4}$, where $\alpha_1 = 10^{-7}$, $\alpha_2 = 10^{-8}$, $\alpha_3 = 10^{-7}$, $\alpha_4 = 10^{-8}$. Mutual Information: PDF(BA Algorithm): 8.147 bits, Jeffreys Prior: 8.087 bits, Data: 8.092 bits

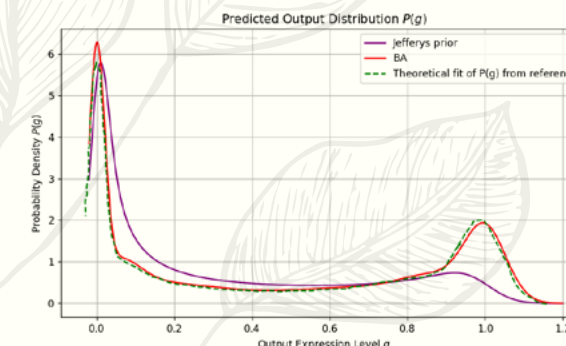


Figure 4: The output distributions $P(g)$ computed from the corresponding optimal input distributions obtained through the BA algorithm and Jeffreys prior in comparison with the result presented in the reference paper. Noise Model: $\sigma_g^2(\bar{g}(c)) = \alpha \bar{g} + \beta \bar{g}^{1.8} (1 - \bar{g})^{2.2} + \delta$, where $\alpha = 2.5 \times 10^{-3}$, $\beta = 0.5$, $\delta = 4 \times 10^{-4}$. Mutual Information: Reference Paper: 1.7 bits, BA Optimal: 1.77 bits, Jeffreys Prior: 1.54 bits

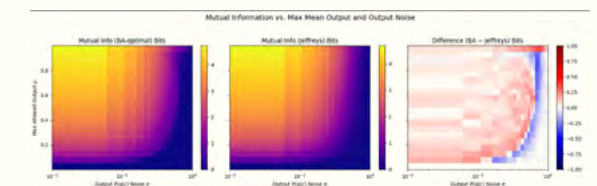


Figure 5: Heat Plots of Mutual Information (Channel capacity) calculated using BA algorithm and Jeffreys Prior

Explanation: In Figure 3, We observe that Jeffreys Prior is matching with the plot generated from data in the paper. But, in Figure 4, Reference data is matching with Jeffreys Prior at lower expression level and with optimal $P(g)$ from BA Algorithm at higher expression level indicating that Jeffreys Prior is more suitable in lower noise whereas BA Algorithm is valid in higher noise.

Figure 5 compares both BA algorithm with Jeffreys Prior and shows that both of these are equally good at lower noise and show a difference of almost ± 0.25 bits at high noise.

4 Keywords

Mutual Information, Fisher Information, Jeffreys Prior, Blahut-Arimoto Algorithm

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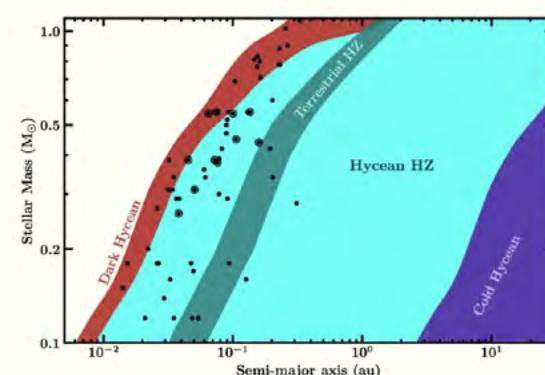
Life in the Infrared

Arya Raj (BSMS'21)

The era of searching for Habitable Worlds is revolutionized by the invention of the James Webb Space Telescope (JWST). The sub-neptune planets including K2-18b are a major investigation of choice. Recent observations through JWST using Mid-Infrared Instrument (MIRI) reveal that the compelling spectral features in the atmosphere of K2-18b point out the presence of dimethyl sulfide (DMS) and dimethyl disulfide (DMDS) which are the important bio-signatures. Here is the brief summary of major findings of Madhusudhan et al. (2025), who present the first mid-infrared transmission spectrum of a habitable-zone exoplanet, and discusses the implications for exoplanetary biosignature detection and the broader astrobiological significance of their results.

The search for biosignatures on exoplanets has been significantly accelerated by the capabilities of the James Webb Space Telescope (JWST), which allows for detailed atmospheric characterization of temperate planets around M-dwarf stars. Among the most promising targets is K2-18 b, a sub-Neptune-sized exoplanet located in the habitable zone of its host star. Recent studies have proposed that K2-18 b may belong to a new class of potentially habitable worlds known

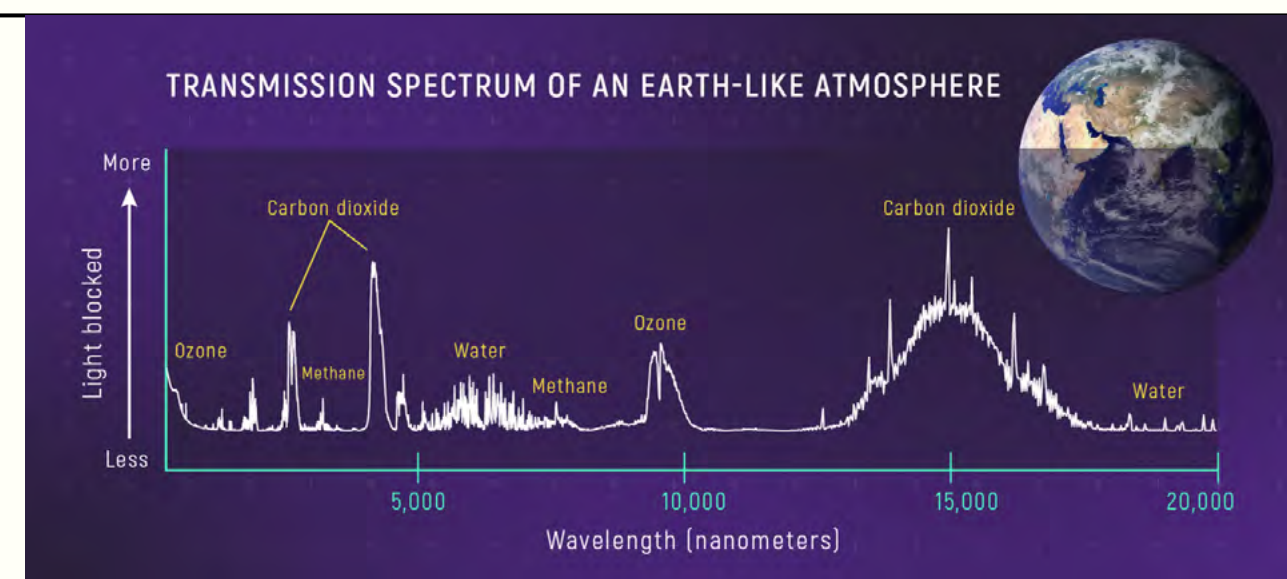
as “hycean planets,” characterized by a hydrogen-rich atmosphere and a global liquid water ocean.



A plot showing how Hycean worlds maintain liquid water over much larger orbital distances than terrestrial planets, significantly widening the search for life.

Courtesy: Nikku Madhusudhan et al

Early near-infrared observations with JWST detected methane (CH_4) and carbon dioxide (CO_2) in its atmosphere - molecules whose simultaneous presence, especially in a shallow H_2 -dominated atmosphere, hints at conditions favorable for life. Building on this initial evidence, Madhusudhan et al. (2025) report the first mid-infrared transmission spectrum of K2-18 b using JWST's MIRI instrument, revealing new spectral features that point toward the presence of dimethyl sulfide (DMS) and dimethyl disulfide (DMDS). Both molecules are considered promising biosignatures on Earth, as they are primarily produced by marine life

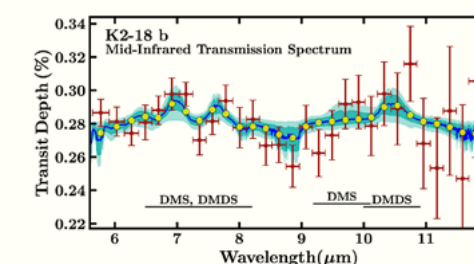


Courtesy: NASA

and are difficult to sustain in large quantities through abiotic processes in H_2 -rich atmospheres. The detection of these gases at $\sim 3\sigma$ significance represents an important development in the search for life beyond Earth, providing independent support for the potential biological activity hinted at in previous observations. This review summarizes the methods, findings, and implications.

The study used the transit method of observing exoplanets. They utilized the Mid-Infrared Instrument (MIRI) on JWST to observe the transmission spectrum of K2-18b during a planetary transit. The observations were carried out in the 6–12 μm range using the MIRI Low-Resolution Spectrograph (LRS) in slitless mode, under JWST GO Program 2722. A total of 5.85 hours of data, including 2.68 hours of in-transit time, were collected to capture atmospheric absorption features. To ensure robustness, two independent data reduction pipelines were employed: JExoRES and JexoPipe. Both pipelines followed JWST's standard calibration steps, but implemented their own approaches for

spectral extraction, cosmic ray correction, and systematic noise removal. The resulting transmission spectra were cross-validated and showed consistent features across both pipelines. Atmospheric retrievals were performed using the AURA framework with Bayesian nested sampling to infer the molecular composition. A wide range of molecules, including known biosignature gases, were tested. The analysis focused on fitting the spectral features with models includ-

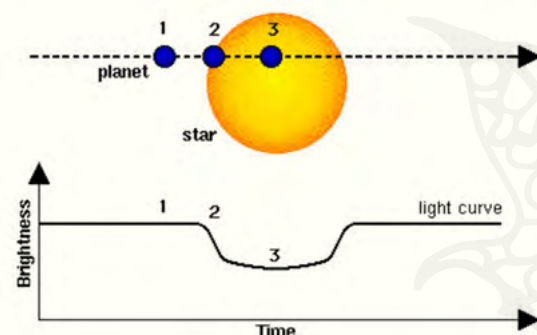


Courtesy: astro-ph.EP

ing dimethyl sulfide (DMS), dimethyl disulfide (DMDS), CH_4 , and CO_2 , while accounting for cloud opacity, temperature profiles, and potential time-correlated noise.

The mid-infrared spectrum of K2-18

b obtained from JWST MIRI showed distinct spectral features between 6 and 11 μm that deviate significantly from a flat, featureless spectrum. These features were not attributable to common atmospheric constituents like CH_4 or CO_2 , which have limited or weak absorption in this range. Instead, the spectral shapes aligned closely with the known absorption features of dimethyl sulfide (DMS) and dimethyl disulfide (DMDS), both of which are sulfur-based molecules with strong mid-infrared signatures.



"How to find an extrasolar planet"
Courtesy: The European Space Agency

Atmospheric retrievals consistently indicated the presence of DMS and/or DMDS at high abundances, with volume mixing ratios ≥ 10 ppmv. Models including either molecule individually yielded detection significance around 3σ , while models including both improved the fit but revealed a degeneracy between them. The retrievals did not find evidence for other potential biosignature gases or cloud opacity in this spectral range. These findings held true across different retrieval setups, binning schemes, and data reduction pipelines, confirming the robustness of the result. While DMS and DMDS are not unambiguous bio-

signatures, their detection is notable because they are primarily produced by biological processes on Earth. Their coexistence with CH_4 and CO_2 , observed in previous JWST near-infrared spectra of K2-18 b, supports the possibility of a biologically active hycean world. However, the authors caution that further observations and improved molecular databases are needed to distinguish between biotic and plausible abiotic sources.

The potential detection of dimethyl sulfide (DMS) and dimethyl disulfide (DMDS) in the atmosphere of K2-18 b carries significant astrobiological implications. On Earth, DMS is produced almost exclusively by marine microorganisms, and both DMS and DMDS are generally considered strong biosignature candidates due to their limited abiotic production pathways. Their presence alongside previously detected methane (CH_4) and carbon dioxide (CO_2) in a hydrogen-rich atmosphere supports theoretical predictions that such combinations may signal biological activity on habitable exoplanets. The inferred abundances of DMS/DMDS (≥ 10 ppmv) are consistent with models assuming biogenic emission rates several times higher than Earth's, especially under the ultraviolet conditions typical of M-dwarf host stars. However, confirming a biological origin remains challenging. Abiotic production routes for sulfur-bearing molecules in H_2 -rich atmospheres are not fully understood and could potentially mimic biosignature signals. Moreover, laboratory cross sections used for molecular identification are based on Earth-like conditions and

may introduce uncertainties when applied to exoplanetary atmospheres. While the findings do not constitute definitive proof of life, they represent one of the strongest cases yet for biosignature detection beyond Earth. These results emphasize the need for further observations, both to improve signal confidence and to explore similar candidate biosignatures in other temperate exoplanet atmospheres.

The study demonstrates strong robustness through the use of two independent data reduction pipelines (JExoRES and JexoPipe), consistent spectral features across different light-curve treatments, and confirmation via various atmospheric retrieval configurations. A leave-one-out cross-validation further showed that the detected spectral signatures are not driven by single data points or local systematics. However, limitations remain - primarily the degeneracy between DMS and DMDS due to overlapping features, and the reliance on molecular cross sections measured under Earth-like conditions, which may not accurately represent behavior in H_2 -rich exoplanetary atmospheres. Additionally, the detection significance ($\sim 3\sigma$) falls short of the 5σ threshold typically required for unambiguous discovery, necessitating follow-up observations for confirmation. Future research should focus on acquiring additional JWST observations of K2-18 b across broader wavelength ranges and with higher signal-to-noise ratios to improve the detection significance and resolve the spectral degeneracy between DMS and DMDS. Laboratory experiments

are also essential to obtain accurate absorption cross sections for potential biosignature gases under H_2 -rich, high-temperature conditions relevant to exoplanetary atmospheres. Comparative studies of other hycean candidates will help determine whether such biosignature signals are unique to K2-18 b or indicative of a broader pattern, thereby refining our understanding of habitability and life beyond Earth.

The mid-infrared transmission spectrum of K2-18 b from JWST MIRI provides compelling evidence for the presence of dimethyl sulfide (DMS) and/or dimethyl disulfide (DMDS), both considered promising biosignature gases. Detected at $\sim 3\sigma$ significance and at relatively high abundances, these molecules present an exciting possibility of biological activity on a temperate sub-Neptune.

While the findings are robust across multiple analyses, limitations such as spectral degeneracies and uncertainties in molecular data prevent definitive conclusions. Abiotic pathways for these gases remain poorly constrained, and further observations are required to confirm their presence and origin. Nevertheless, this study marks a significant milestone in exoplanetary science by demonstrating that biosignature detection is now within reach. It highlights the transformative potential of JWST and sets the stage for a new era of atmospheric characterization and the search for life beyond earth.



THE EVENT HORIZON: LOOKING BACK WITH DR. NITIN YADAV

Currently at IIT DELHI

Your research prominently features magnetohydrodynamics (MHD) theory and simulations. Could you explain how MHD principles help us understand the behavior of the solar atmosphere?

Magnetohydrodynamics combines fluid dynamics with Maxwell's equations to model ionized gases (plasmas) as conducting fluids. Since over 99% of the visible Universe is plasma, we can treat regions of the Sun - such as the photosphere, chromosphere, and corona - as fluids influenced by magnetic and electric fields. To perform an MHD simulation, we: Define the numerical domain (spatial and temporal scales). Set initial conditions based on observations (e.g. active regions for flares). Solve the coupled MHD equations to track the evolution of plasma and magnetic fields over time.

This framework lets us study how perturbations - such as waves excited by turbulent convection - propagate, interact with magnetic structures, and potentially heat the solar atmosphere.

In one of your studies, you explored wave propagation near coronal null points and developed a wave-mode decomposition technique. What challenges do null point regions present, and how does your approach improve our understanding of wave behavior and energy dynamics there?

Magnetic null points - locations where the field strength falls to zero - are ubiquitous in the corona and critical for particle acceleration and

heating. Waves encountering a null point refract and accumulate there, which enhances local dissipation. Existing decomposition methods were limited to 2D or highly symmetric flux tubes. I developed a three-dimensional technique that: Traces individual magnetic field lines throughout the simulation volume. Uses the local geometry of each field line to separate perturbations into the three MHD wave modes (fast magnetoacoustic, slow magnetoacoustic, and Alfvén).

This allows us to track how each mode interacts with nulls - showing, for example, that fast magnetoacoustic waves become trapped at nulls and dissipate into heat.

Could you describe the computational methods and models you use for simulating wave propagation and turbulence in solar plasma?

I employ two main MHD codes: Finite-volume codes, which are inherently more diffusive and handle shocks robustly. Finite-difference codes with explicit numerical diffusion terms.

Key considerations include: Physics fidelity: Incorporating realistic equations of state, energy sources and sinks (e.g., radiative losses), and appropriate transport coefficients. Scale separation: Balancing large-scale dynamics (coronal loops, active regions) with small-scale features (turbulent cascade down to kinetic scales).

Ultimately, we choose the numerical scheme and physics modules based on the specific solar

phenomenon under investigation.

What initially inspired you to pursue physics - and specifically solar physics?

My undergraduate mentor encouraged me to study physics for broader career options, even though I was initially drawn to pure mathematics. During my master's, I worked on laser-induced fusion but found limited opportunities. A subsequent engineering degree project on space plasma sparked my interest in heliophysics, leading me to pursue a PhD on coronal heating and, eventually, postdoctoral work on solar MHD.

What broader implications does your research have for understanding solar energetic events such as jets and flares?

Magnetic reconnection at both large and small scales (including nanoflares) is central to coronal heating and dynamic phenomena like jets. My work on wave trapping at null points complements reconnection-based heating, suggesting multiple, pervasive sites where wave energy can convert into thermal and kinetic energy - helping to accelerate the solar wind and drive jets.

You joined IISER Thiruvananthapuram in February 2022. How has your experience been as a researcher, principal investigator, and educator?

Leading my own projects has been immensely rewarding. Designing research questions, mentoring students, and securing funding have allowed me to pursue ideas born from years of simulation work. I also strive to foster an open, collaborative environment - encouraging students to share and challenge ideas freely, as I experienced during my postdoctoral appointments abroad.

What advice would you give to students aspiring to enter this field?

Build strong fundamentals in classical mechanics, electrodynamics, fluid mechanics, and plasma physics. Develop coding expertise: Learn not just syntax, but problem-solving approaches for modeling physical systems. Seek mentorship and collaboration: Discuss your ideas openly with seniors and faculty to refine your perspective. Maintain balance: Pursue interests outside research - sports, music, or time with family - to stay motivated and avoid burnout.



Dr. Nitin with her students



Dr. Nitin with the Interviewers



EVOLVING PATTERNS: AN INTERVIEW WITH **DR. CHANDRAKALA MEENA**

Currently at IISER PUNE

To begin with, your work in nonlinear dynamics, pattern formation, and complex systems has inspired many students. What initially drew you to this field?

Actually, my interest developed gradually during my studies. In the first two years, we had to study many subjects, and I found myself enjoying physics, mathematics, biology, and chemistry alike. Later, when I took a course in nonlinear dynamics during the fourth semester, I realised that this field allows us to connect ideas from many disciplines. We can model real-world systems, study their behaviour, and understand how different components interact. That is what attracted me to this area, and I continued in it after that.

What are some of the fundamental theoretical challenges in studying complex networks and complex systems?

One important challenge is the classical stability-diversity paradox. Ecology often suggests that a more diverse and complex system should be more stable, but theoretical results show that large and complex systems may actually become less stable. This has been a long-standing puzzle since the 1970s. Many researchers have studied it, but most work focused only on network structure and did not fully include the effect of dynamics. In our work, we tried to combine both network topology and dynamics to understand stability more realistically.

What are the most significant problems you currently study in this area?

My work mainly focuses on stability in complex systems. A system may settle at a fixed point, but that is not always the case. Sometimes the system reaches a periodic state, so we also need to understand the stability of periodic solutions. Another challenge is higher-order interaction. In many systems, components do not interact only in pairs; they interact in groups. For example, in the brain, many neurons participate simultaneously. Including these group interactions makes the analysis much more difficult, but also much more realistic.

How important is collaboration in your research?

Collaboration is essential. My background is in nonlinear dynamical systems, but real problems usually require knowledge from several areas. For example, if we want to study biological systems, we must first understand the biology itself. Then we need ideas from network science, dynamics, differential equations, linear algebra, and numerical methods. Working with people from different fields helps us build a better model and solve the problem more effectively.

Could you share an example of a successful collaboration?

Yes. After joining IISER Thiruvananthapuram, I collaborated with Professor Ambika. In that work, we combined recurrence measures with machine learning. I had experience in machine learning, while she specialised in extracting recurrence features from time series. By combining both approaches,

we published four papers. That collaboration was very valuable for me.

What kinds of emerging phenomena do you try to understand in your work?

In complex systems, the behaviour is often irregular, heterogeneous, and sometimes chaotic. One challenge is to identify whether a system is chaotic and, if it is, how to control it. We also try to understand how to model such systems properly: which components are connected, what interactions they follow, and how we can represent microscopic behaviour through equations and simulations. The systems are usually large, and computationally handling them is also difficult.

How do you approach this challenge in practice?

We usually try to understand the macroscopic behaviour by making reasonable assumptions. We may not be able to capture everything, but we can still gain useful insight into a complex system through a dynamical framework and numerical simulation. This is especially helpful when studying biological or ecological systems, where the real behaviour is complicated but important patterns can still be discovered.

Your work also seems to be connected with biology and neuroscience. How does interdisciplinary research help there?

It helps a great deal. For example, when we try to understand how the brain works, no single discipline is enough. If we combine knowledge from networks, dynamics, biology, and modern computational methods, we can build a more meaningful picture of the system. The same is true for climate-related problems and ecological systems. In climate science, for instance, we can use dynamics, machine learning, and network science to forecast extreme events or study the effects of temperature fluctuations on species and ecosystems.

Could you mention another application area where your ideas are useful?

One application that I find very interesting is gene regulatory networks. I discussed this with a colleague from the Biology Department who had a large amount of gene expression data but wanted help constructing the corresponding network. We tried to reconstruct the gene regulatory network from the available data so that it could be used in further biological studies. This is a good example

of how mathematics and data-driven methods can support real scientific work.

What kind of background do students need if they want to work in this field?

First, they should understand the basic ideas of dynamics and networks. If someone wants to model social, biological, or chemical systems, they need to know how components interact and how those interactions are represented. For the theoretical side, courses in mathematical methods are very useful. For the computational side, numerical methods and programming are important. If students develop a strong foundation in these areas, they can work on many kinds of complex problems.

How has your research group used machine learning in recent years?

We have explored several directions. In one project, a student worked on identifying the dynamical behaviour of a system only from recurrence plots. The idea was to classify whether the underlying system is periodic, hyperbolic, or noisy. In another project, a student was working on extracting dynamical importance using machine learning, especially when analytical methods are difficult. Both projects are exciting because they combine data-driven methods with dynamical systems theory.

What do you feel when your students achieve something notable?

I feel very proud. My students are working in different directions, and they keep producing new results. Some of them have also started attending conferences, and two of my students even received best poster awards. It is always a wonderful feeling to see them grow and perform well.

Your research also deals with pattern formation. What inspires that part of your work?

Nature is one of the biggest sources of inspiration. For example, one of my collaborators studied how bacteria grow on surfaces. The resulting patterns depend on the surface properties and flow conditions. Experimentally, this is expensive and difficult to study in detail. But numerically, we can model the situation with differential equations, tune the parameters, and predict which conditions will produce which patterns. That makes simulation a powerful tool.

How do you view the role of artificial intelligence and machine learning in the future of this field?

AI and machine learning are very useful, but they do not replace human understanding. A machine can only do what it has been trained to do, and it still needs human guidance. At the same time, these tools can help us explore real-world systems, extract hidden dynamics, and study connections that are difficult to capture analytically. I think they will become even more useful in future research, especially when combined with dynamical systems and network science.

What is your opinion on the changes in research and learning tools over time?

Things have become much easier in many ways. Earlier, we had to write numerical codes from scratch, but now many libraries and integrators are available in Python and other platforms. There are also many online resources, lecture videos, and courses such as NPTEL. Even tools like ChatGPT can be helpful for generating ideas or hints, as long as one does not rely on them blindly. We still need critical thinking, but the availability of resources has definitely improved the learning process.

How has your experience been here at IISER TVM?

I have enjoyed my time here a lot. The campus is beautiful and peaceful, and many facilities are available here. The academic and cultural environment is lively, and events such as the Frontier Symposium are very engaging. I especially like the short five-minute talks, where people explain their work briefly and clearly. I also enjoy the cultural atmosphere, including the festivals and the local traditions.

Finally, what message would you like to give to students interested in physics and complex systems?

My advice is to build a strong foundation in mathematics, especially in theoretical and numerical methods. If you understand the basics well, you can move into many directions, whether in physics, biology, data science, or network science. This field is wide, interesting, and full of open problems. If students stay curious and patient, they can contribute meaningfully to it.

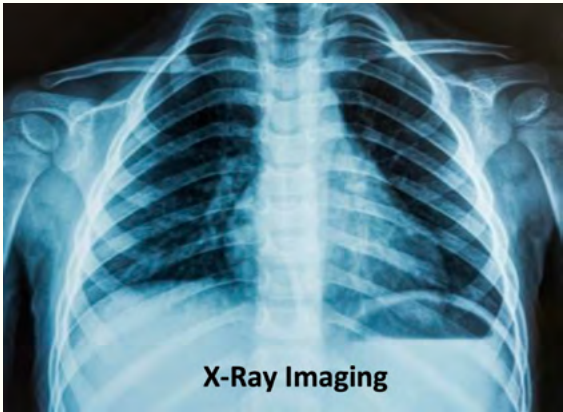


Dr. Meena with the interviewers

SEEING BENEATH THE SKIN
THE PHYSICS AND PROMISE OF
PHOTOACOUSTIC IMAGING

Aranya Sarkar (PhD)

Have you ever wondered how physicians understand the human body so well? Or, how do they find out about our ailments just by looking at a piece of paper showing some black and white or colourful images? There



comes the enchanting world of biomedical imaging. Modern medicine owes much of its diagnostic power to physics, from the sharp silhouettes of X-ray images to the mesmerising depth of MRI scans. Biomedical imaging - at the intersection of electromagnetic waves, sound, and computation - has long allowed us to peer beneath the body's surface without making a single incision. Biomedical imaging isn't a monolith but a spectrum of techniques, each grounded in physics and optimised for specific contrasts. The biomedical imaging revolution kick-started with the discovery

of X-rays by Röntgen in 1895. X-ray radiography relies on differential attenuation of high-energy photons governed by bulk tissue properties. It was followed by the discovery of Nuclear Magnetic Resonance (NMR) in 1946 by Felix Bloch and Edward Purcell, and the successive introduction of Magnetic Resonance Imaging (MRI). MRI depends on the nuclear spin precession of protons or Hydrogen nuclei in our tissues and their successive excitation and relaxation to RF pulses. Finally, it was followed by medical ultrasound imaging in the 1950s-60s, which varied with a tissue's acoustic wave propagation and acoustic properties. These are not just tools - they are experiments running quietly inside the body, facilitating an insight into various anatomical and patho-physiological changes for the prognosis of disease conditions. Each modality offers its strengths and tradeoffs.

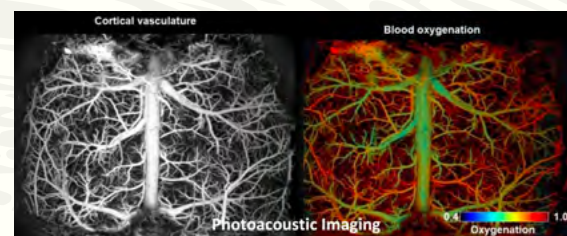
Modality	Physical Principle	Key Strength	Main Tradeoff
X-Ray	Photon Attenuation	High bone contrast	Ionizing radiation
MRI	Nuclear Spin Relaxation	Soft tissue detail	Expensive / Slow
Ultrasound	Acoustic Reflection	Real-time / Portable	Lower resolution
Photoacoustic	Light-to-Sound Conversion	Optical contrast + Depth	Depth-limited by light

X-rays are fast and cheap but involve

ionising radiation, which harms our bodies. MRI offers excellent soft tissue contrast, but at a high cost and slow acquisition times. Ultrasound is realtime and portable, though limited in resolution and contrast. And optical imaging, while rich in biochemical sensitivity, is stymied by light scattering in tissue, restricting it to superficial imaging - until photoacoustics entered the scene.

From the last decade of the 20th century, researchers and clinicians started introducing multi-modal imaging in healthcare, fusing different contrasting imaging technologies to develop a beautiful synergy, which can give us structural and functional information about our bodies. They started researching heavily on artificial intelligence for improved image quality and achieving greater depths and resolutions. Photoacoustic imaging (PAI) caught the attention of physicists, engineers, and clinicians at this juncture. First observed by Alexander Graham Bell in 1880, the clinical use of photoacoustic imaging came much later with the advent of sensitive ultrasonic detectors and high-energy and precision pulsed lasers. At its core, PAI offers a beautiful hybrid of optics and acoustics. It exploits the fact that light is absorbed and converted into heat, producing pressure waves we can detect as sound. This photothermal-to-acoustic conversion allows us to sidestep the significant limitation of optical imaging - scattering in tissue - while retaining the rich contrast that optical absorption offers, simultaneously achieving greater depths of ultrasound imag-

ing. Another powerful side of PAI is its noninvasive nature, making it safer to use than X-ray or other radiation imaging modalities and cheaper than MRI. On top of that, it can act as both a structural and functional imaging technology.

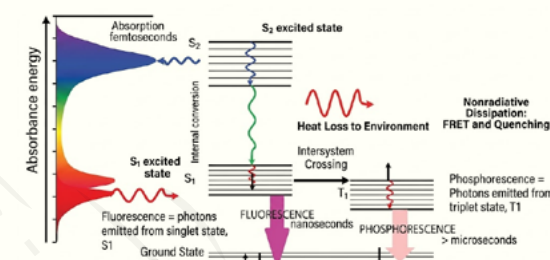


Courtesy: Duke University

The principle behind photoacoustic imaging is surprisingly intuitive: shine a short laser pulse onto a tissue, and wherever light is absorbed - say, by haemoglobin or melanin - the tissue heats up slightly, expands, and launches a pressure wave.

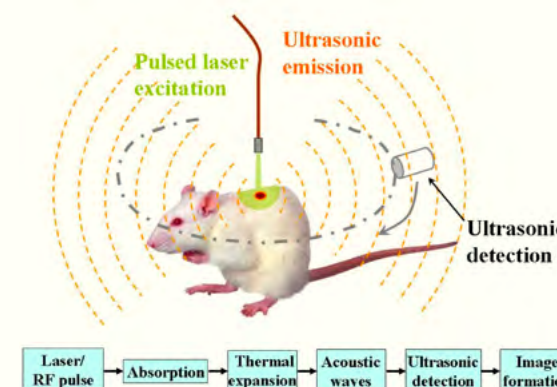
This wave, travelling at the speed of sound, is then captured by ultrasound detectors. While the thermal expansion is minuscule, the resulting acoustic signal is strong enough to be detected with high fidelity. And because sound scatters much less than light in biological tissues, the resolution and depth of imaging are significantly improved. Mathematically, the pressure signal is proportional to the pulsed laser light source's energy deposition rate and the Grüneisen parameter. This thermo-mechanical coefficient relates absorbed energy to pressure generation. The pressure signal is collected as an electric signal via the piezoelectric transducers, and successive signal processing lets you reconstruct optical absorption maps

- a form of acoustic tomography with optical contrast.



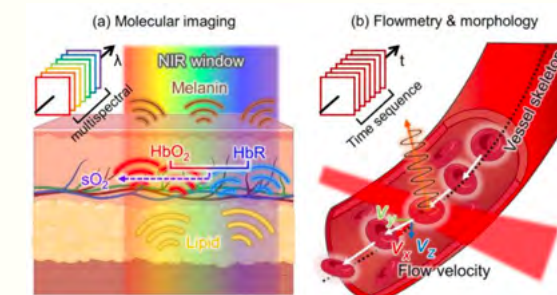
The Jablonski Diagram of molecular absorbance and fluorescence.

If you delve deeper into the atomic theory, you will understand photoacoustics more clearly. Whenever an atomic transition happens due to the absorption of electromagnetic energy, energy is released from the atom.



Schematic illustration of photoacoustic imaging.
Courtesy: Wikipedia

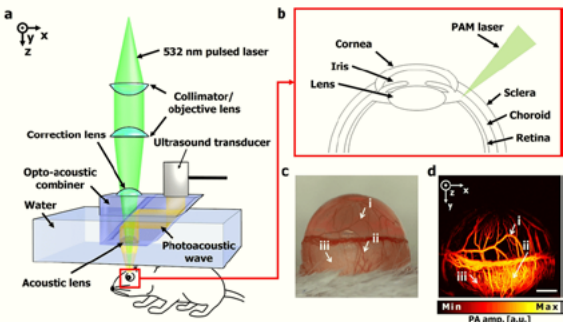
In this case, we are talking about molecular energy bands, which have rotovibrational states between each band. Whenever there is a nonradiative fast transition in between roto-vibrational energy states inside a high-energy electronic molecular band, a phonon, which is a particle responsible for thermal energy transfer, is released, which has energy in the infrared region, creating thermal



Courtesy: ResearchGate

expansions in the tissue of interest. Again, a fluorescence photon is released if a radiative transition happens between a higher and a lower energy band. These two processes compete with each other for a sample of interest. They also depend on the transition time interval. Usually, non-radiative phonon transitions occur at a much smaller time interval of nanoseconds, while the other occurs in microseconds. Thus, we need a nanosecond pulsed laser for the photoacoustic effect in a biological tissue with molecules with near-infrared energy bands. Haemoglobin, the oxygen-carrying molecule in blood, is a strong absorber of near-infrared light. Also, Melanin, a chromophore responsible for humans' skin colour, and Cytochrome, an enzyme in mitochondria, responsible for electron transport during cellular respiration, are a few of the vital chromophores in a tissue. With PAI, these molecules serve as endogenous contrast agents, meaning we don't need to inject anything artificial to see them. One can even distinguish between oxyhemoglobin and deoxyhemoglobin by imaging at multiple wavelengths, enabling real-time mapping of blood oxygenation - functional information crucial in oncology, neurology, and cardiovascular research.

Cytochrome concentrations depict the oxidative states of a cell, whether it's hypoxic or normoxic. These are essential biomarkers for understanding tissue tumour environment and detecting circulating cancer cells. This dual capability - structural and functional imaging - makes PAI a diagnostic tool and a window into dynamic physiology. In small animal models, photoacoustics has already



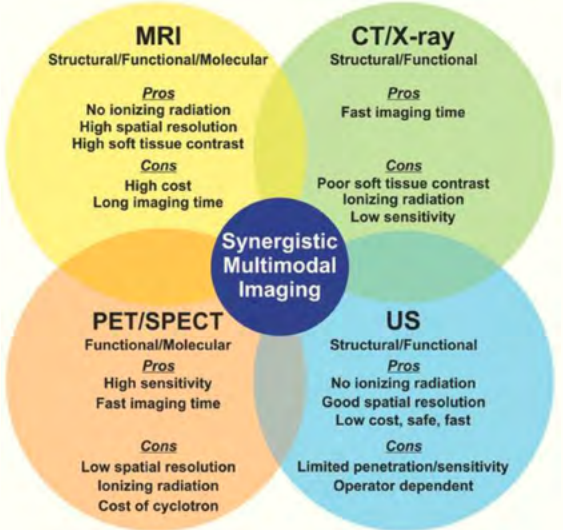
A comparison of Photoacoustic Tomography (PAT), which captures whole-organ structures at depth, and Photoacoustic Microscopy (PAM), which provides sub-micron detail of individual microvessels. Courtesy: ResearchGate

been used to image brain activity via cerebral cytochrome uptake through the skull or track tumour growth by visualising angiogenesis using oxygen saturation maps.

There are several flavours of PAI, depending on the scale and application. In photoacoustic tomography (PAT), the laser illuminates a broad area, and an array of ultrasonic unfocused detectors is used to reconstruct a 3D image.

This is well-suited for imaging organs or tumours a few centimetres deep with sub-millimetre resolution. On the other hand, photoacoustic microscopy (PAM) achieves sub-micron

resolution by tightly focusing the light or sound, making it ideal for imaging capillaries inside blood vessels or even individual cells in superficial layers. The laser fluence and energy are varied to make it safer for live animal imaging. Another parameter that can be varied is the transducer detection bandwidth. The higher the central frequency value, the lower the depth of the sound signal, and the higher the resolution. Megahertz transducers are mainly used in PAI. For example, a 5 MHz transducer has a few centimetres depth with very low resolution compared to a 50 MHz transducer with a shallower depth of millimetres with very high resolution. Beam-forming algorithms and noise-based data filtering rectify these drawbacks. Techniques include delay-and-sum algorithms, synthetic aperture focusing technique, backprojection, and more advanced model-based inversions considering acoustic heterogeneity or fluence variation along the laser illumination volume. There's growing interest in deep learning models that can



Courtesy: Jonathan F Lovell via ResearchGate

accelerate or even improve these reconstructions.

While photoacoustics shows immense promise, it's not without challenges. Much research is being done on beam-forming algorithms and reconstruction methods to achieve better precision and accuracy of PAI. This can be a niche work area for computational and experimental physics enthusiasts. Quantitative imaging, on the other hand, remains a significant hurdle. Because the signal depends on absorption and local light fluence (which varies due to scattering after crossing a few millimetres of tissue layers), accurately estimating chromophore concentrations can be tricky. Moreover, translating PAI to the clinic requires solving practical issues, such as integrating lasers and ultrasound transducers into compact, user-friendly devices. But significant strides have already been made. Handheld PAI systems now exist, and combined ultrasoundphotoacoustic probes are entering clinical trials for breast and thyroid imaging. Another exciting frontier is theranostics - combining diagnosis and therapy. Photoacoustics can guide laser ablation, track drug delivery using contrast agents, or even monitor the effects of photothermal therapy in real time. In a sense, PAI is still growing into its full potential. As it integrates with other gold-standard clinical diagnostic imaging modalities - MRI, CT, PET - or with wearable tech, we may find ourselves routinely using it to monitor health, diagnose disease, or guide therapies. The fact that a pulse of light can be turned into a

proper acoustic signal that reveals biological function is a testament to the ingenuity of applying classical physics in modern contexts. And as lasers get faster, transducers become more sensitive, and reconstruction algorithms become more robust, photoacoustics is poised to become a mainstay of medical imaging. But beyond its utility, PAI reminds us of something more fundamental: that even in the most complex systems, physical laws - like how light is absorbed or sound travels - can help us see what was once invisible. By synergistically combining the optical and the acoustic domains, photoacoustic imaging is a prime example of how physics continues to shape the future of medicine - one pulse, one wave, one image at a time.

solution
the crossword

a crossword by
Soumyadip Niyogi & Pranav Gupta

Across	
fusion	2
dirac	4
angstrom	7
qubit	10
impulse	11
ohm	13
uncertainty	15

Down	
singularity	1
quark	3
attenuation	5
hamiltonian	6
viscosity	8
brillouin	9
series	12
matrix	14

LED there be Voltage!

Srushti Dessai (BSMS'23) & Satwik Bhat (BSMS'23)

Let me take you through what it's like to be an Earth Science major. In the first observational techniques class, our professor, Anand Sir told us that we had to make a handheld sun photometer as our course project, and utterly clueless faces stared back at him. What would go through your mind when hearing "sun photometer"? Some of you thought of the Glucon-D ad going in reverse. But our brains collectively thought of an advanced device with a high-level microchip, some intricate circuitry of wires and LEDs, something which would require immense funding, and something more fit for an engineering lab than an EES course. Let's first understand what a sun-photometer is: it is a device that measures the intensity of sunlight at different wavelengths. It primarily measures atmospheric aerosol optical depth and understands atmospheric turbidity. As a group of five, we started our perilous journey into engineering land. After some research, we found out what components we would require. The initial plan was to have one component to detect solar radiation, one to measure it, one to display and then a data acquisition. A spectrally selective photometer requires a bandpass filter. However, we quickly realised the difficulty of working with one in an initial prototype. They are also relatively expensive. Thus, we opted for a jankier solution. We had come across various papers that discussed how a simple light source could be used as

a light detector. As the alternative relied on expensive and problematic filters, we decided to test out this oxymoron: using an LED. Not only are LEDs budget-friendly, but they are also available in various colours, making them spectrally selective. We presented this stroke of genius to our guide, and to our utter surprise, he was on board. He asked us to try it out and observe for ourselves.

Naturally, we decided to discuss this with seniors to check whether our absurd idea had a shot at working, but all we got were puzzled reactions and general disbelief. There was only one thing left to do. We got a few LEDs, a multimeter and a few wires to conjure up a truly DIY looking device. We checked whether we got an output when the light intensity changed, and there it was. The oxymoron had worked and given us a working detector. We had our project, all we needed to do was make it spectrally selective and then test if we could get the Beer-Lambert law. The idea was that when an LED emits light in a particular band or colour, it should also absorb light in a similar band and produce a voltage signal, making it an intrinsic filter. To cover the visible spectrum, we used four different LED colours, red, yellow, green, and blue.

After setting up the circuit on a breadboard, we faced another obstacle: blocking diffused light. This was critical as our photometer needed to

measure direct solar radiation. We decided to stick to typical Indian ingenuity and made a cardboard box with cutouts for black PVC pipes to go into, blocking off all the unnecessary light from reaching the LEDs. We planned to take readings across a whole day to observe the diurnal variation of solar intensity. The terrace of SEESS block was chosen as our observation site.

We began early in the morning, setting up our unpretentious setup right on the terrace floor. Armed with just a multimeter and some recording material, we took readings every half hour to maintain the consistency. The sun blazed down as we crouched beside our cardboard-and-PVC setup, watching numbers fluctuate on the multimeter. The values began to rise and reached their peak by noon and decreased thereafter. The highest voltage was found for green and the lowest for blue LED, and the readings were in millivolts, as we expected.

To check for atmospheric turbidity, the readings had to be taken with a clear sky, but our solar streak was cut short by clouds. Once we plotted the readings for the time of day on a logarithmic scale, we were surprised to find that it closely resembled a Langley plot. We had unpacked a result grounded in actual physics. We repeated the experiment a few days later, but this time the skies weren't as supportive. Clouds came and went, forcing us to stay patient and take readings at irregular time intervals, but we still managed to gather enough data for the report.

What followed was utterly unexpected. We explained our method and voltage curves to the professor. He

looked at it, nodded, and approved of our method, only suggesting minor corrections. That was the real surprise of this project: not the success of the experiment, but how such a toy-like setup, made of LEDs, wires, inside a scrap-like box, some PVC pipes and a multimeter, had passed the test with flying colours.



Figure : Setup of the Experiment

Looking back, that whole experience was one of the highlights of our semester. It wasn't just about getting the marks for that project; it was about learning to test hypotheses, to improve upon ideas and work in a group. The weekly presentations, the meetings, long terrace hours, and the thrill of seeing a real Langley plot emerge from our crude setup; it all made the learning more real and meaningful. The professors were approachable and encouraging. It was our first encounter with experimental science, which left a lasting impression. In the end, it did not matter that our device looked janky or unfinished. It worked, and that is all that matters.



FIRST LIGHTS

Anay Mishra (BSMS'24)

One of the most mysterious and exciting phenomena in our physical universe is light. Light has inspired scientists for centuries and continues to play a vital role in modern optical technology. But what if we could trace the path of distant light back in time to reveal insights into the past and provide clues for our future? Though the light from the first stars and galaxies provides a window into the “cosmic dawn” which is estimated to have occurred around

800 million years after the Big Bang, an even more special kind of ancient radiation, known as the ‘First Light,’ was formed about 380,000 years after the Big Bang, when the universe had cooled enough for light to travel freely. In science, this radiation is known as the Cosmic Microwave Background (CMB). The CMB opens humanity’s window into the most ancient chapters of our universe.

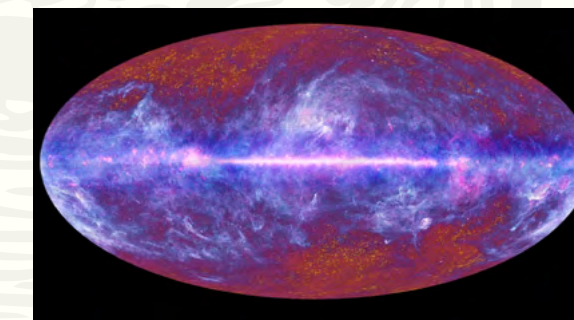
This artist's impression shows a field of Population III stars as they would have appeared a mere 100 million years after the Big Bang. Astronomers may have discovered the first signs of their ancient chemical remains in the clouds surrounding one of the most distant quasars ever detected.
Courtesy: NOIRLab

The universe is thought to have begun in darkness. It started as an extremely hot and dense point, so powerful that it created nuclear particles within minutes, the moment when matter first formed in our universe. Though mysterious and quiet, the Dark Ages carried the potential for the future. During this prolonged age, the universe continued to cool, leading to the creation of the first lighter elements such as hydrogen and helium in an event known as recombination. As the universe became transparent to light, it released the CMB. With no stars or galaxies formed yet, this faint microwave background is visible in the spectrum as a remnant of that early time. Meanwhile, tiny density fluctuations grew under gravity, becoming the seeds of the first cosmic structures. Early gas clouds collapsed under gravity, with starlight from the first stars ionizing the neutral hydrogen henceforth marking the end of the Dark Ages.

The Cosmic Microwave Background, our ultimate tool for peering into the past, was famously detected by Arno Penzias and Robert Wilson at Bell Labs in 1965 (the prediction came earlier, in 1948, by theorists like George Gamow). In the early universe, countless photons were scattered by free electrons, making the universe opaque. As atoms formed, free electrons combined with nuclei, and photons could travel freely. This last scattering is the origin of the CMB. As these photons have traveled through expanding space ever since, they have been stretched into microwave wavelengths due to cosmic redshift. Although faint, the CMB is incredibly powerful for understanding the conditions of the young universe (just 380,000 years old), a blink in the

cosmic timeline.

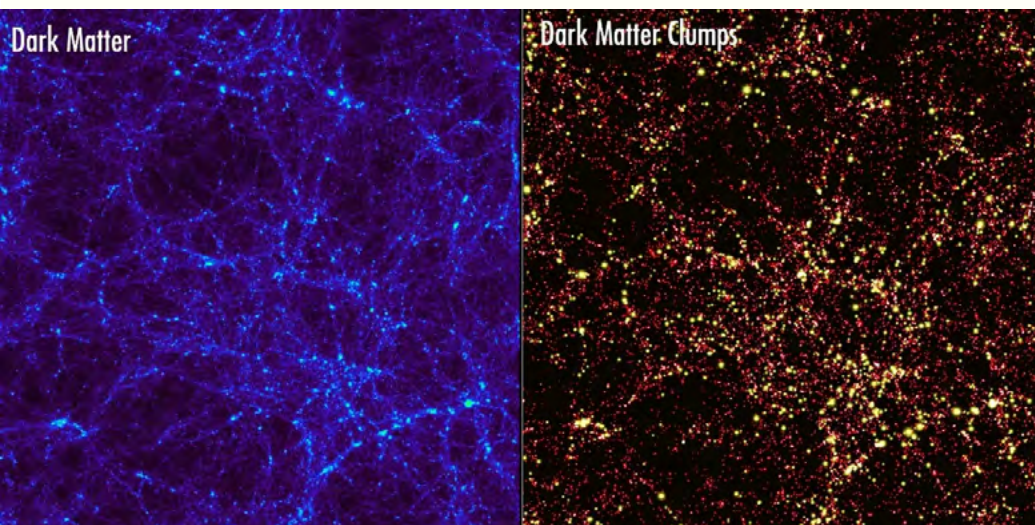
Temperature fluctuations show how matter density varied across space, seeding the formation of early galaxies and large-scale structures. The CMB also provides strong evidence that our universe is mostly made up of dark matter and dark energy, and it is con-



All-sky image of the cosmic microwave background
Courtesy: (ESA/ LFI & HFI Consortia)

sistent with the flat universe model, a foundation for modern cosmology. It allows precise estimates of the universe’s age (about 13.8 billion years). The CMB’s subtle patterns fit well with inflation theory, which suggests that the universe expanded faster than the speed of light immediately after the Big Bang. Its polarized nature reveals information about the Epoch of Reionization, and recent research aims to find hints of primordial gravitational waves within it.

Among the early cosmic sources, Population III stars are especially significant. Classified by their metal content, these were the first generation of stars, formed from primordial gas clouds made almost entirely of hydrogen and helium. Lacking heavier elements, they were massive (about 10 - 1000 times the mass of our Sun) and had short lifespans, lasting only a few million years. Though we have



Simulations of dark matter distribution in the Universe during early galaxy formation period
Courtesy: Virgo consortium

never seen them directly, their chemical fingerprints are found in ancient, metal-poor stars and intergalactic gas. When these massive stars died in powerful supernova explosions, they dispersed newly formed heavy elements into the interstellar medium. These elements later became building blocks for the next generations of stars, gas clouds, planets and eventually life itself. These first stars were so intense that they ionized surrounding neutral hydrogen gas, turning it into plasma and creating the transparent universe we observe today allowing us to study the CMB and unlock secrets of the infant cosmos.

Dark matter, though invisible, makes up most of our universe's mass. Unaffected by radiation after the Big Bang, it clumped together to form gravitational wells called dark matter halos. These halos pulled in gas clouds, which cooled and condensed to form clusters of stars known as proto-galaxies. As halos merged, small star clusters combined to form small galaxies, which then merged into larger galaxies and clusters. The very first galaxies were small, irregular, and clumpy,

often merging violently and triggering intense bursts of star formation. Major CMB experiments have mapped the tiny temperature fluctuations (anisotropies) in exquisite detail that reveal the seeds of early galaxies. Modern galaxy surveys and powerful telescopes look deep into space and back in time to study this

light from early galaxies, giving us priceless insights into cosmic evolution.

As we step into a golden age of physics and astronomy, light from the universe's origins connects us to the cosmos in a profound way. With powerful observatories like the James Webb Space Telescope, we gain new insights into nature, from microscopic atoms to vast galactic structures reminding us that we are made from the same cosmic material. Next-generation researchers aim to probe even closer to the universe's first moments after the Big Bang. When distant, ancient stars die, they spread the elements of life into space, creating conditions that made life on Earth possible long ago. As ancient photons from the early universe reach our eyes, we experience a moment of cosmic intimacy. Light will continue to guide our quest to understand the origin of everything around us.

BLACK HOLE

An Astounding Mystery

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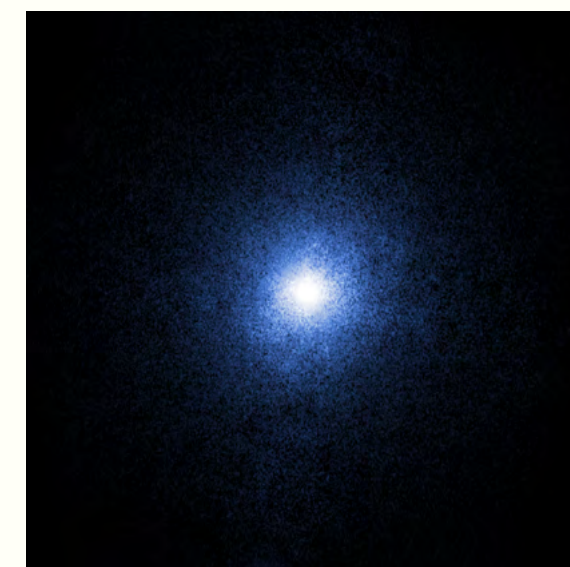
Among the countless mysteries scattered across the universe, black holes remain the most fascinating and terrifying objects ever discovered. Invisible to the naked eye yet powerful enough to trap even light, black holes challenge our understanding of space, time, and gravity itself. This article explores the discovery, current and future road to the research of these astonishing cosmic objects.

The road to the discovery of black holes began with Einstein's Theory of General and Special Relativity. Schwarzschild solved the "unsolvable" and got a weird solution to this crudely approximate theory: a curve which hollowed down to an infinite distance (mathematically to a point of "singularity", a point with infinite density), such that even light can't escape from it once it entered inside. This was too good to be true until it was observed much later.

As science progressed, the study of the atom became prominent. In the late 1930s, Dr. Chandrasekhar and Dr. Oppenheimer proposed a theory of black hole formation, in which it was stated that after a certain mass of a massive star was attained (upon nuclear fusion), the force of gravity

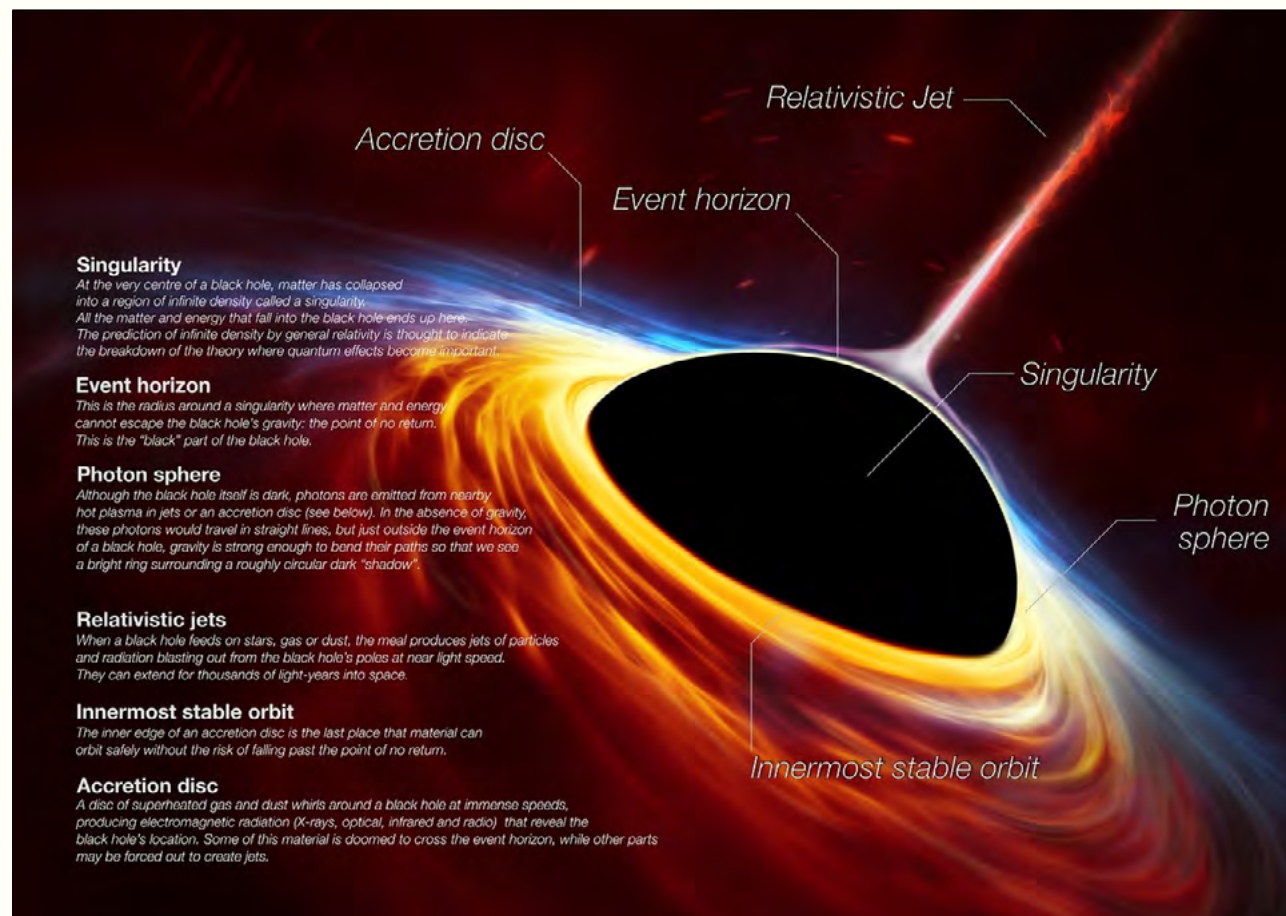
overpowers the pressure built up due to nuclear fusion. Thus, a black hole forms along with a supernova explosion.

The story does not end here. So keep your eyes held onto this article.



Chandra X-ray Observatory image of Cygnus X-1

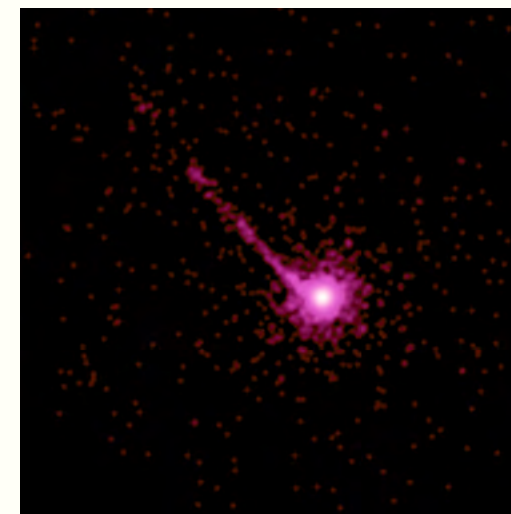
In the 1960s, Paul Murdin, then an astronomer at the Royal Observatory, in England, discovered Cygnus X1, a binary system, consisting of a star and a black hole. Doppler shift was used in this discovery. The system was emitting X-rays. The star of this system orbited around its companion black hole every 5.6 days. The accretion disk around the black hole



This artist's impression depicts a rapidly spinning supermassive black hole surrounded by an accretion disc.
Courtesy: ESO

black holes at the centers of numerous galaxies. These supermassives are formed by direct collapse, i.e. by

Some scientists are speculating about the thought of black holes being essential to the existence of this growing universe.



The Chandra X-ray image of PKS 1127-145, a quasar about 10 billion light-years from Earth.

As black hole research is growing, lots of black hole observations are taken by the James Webb Space Telescope and the Event Horizon Telescope in order to get a better understanding about a variety of black holes. Gravitational waves have become the most important tools to understand and detect black holes, compact binary coalescences and so on. These waves are detected by the LIGO observatories in this world.

sucking in gas clouds or by gulping huge stars in the same way as stated in this article. These supermassives are "necessary evils" for organized structures to exist in our universe.

Though there has been tremendous progress in black hole research, there is a lot left to discover about these astoundingly mysterious objects. This universe will remain to be the most complex and mysterious thing to understand forever.

had particles of the companion star orbiting the hole at half the speed of light. Upon inter-particular collision, these particles emit X-rays. This black hole also had thin jets coming out of the center on both sides.

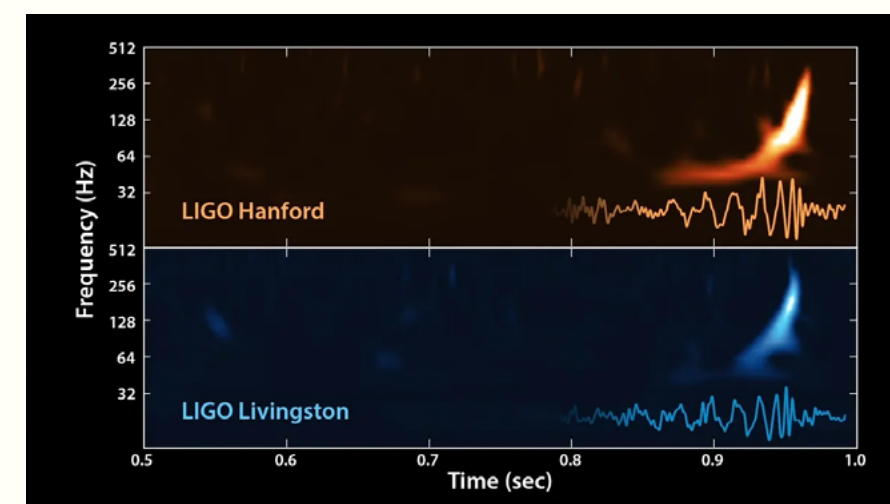
Though a considerable amount of progress was done in this subject, a lot more is awaited in the years to come! Let's look at the structure of a black hole: The black hole mainly consists of an accretion disk, where the material coming under the gravity of the black hole orbits at high speeds, and the event horizon.

As you move towards a black hole, your movement slows down as per the surrounding, but for you, time is going on in the same way as before. You may not be aware when you

have crossed the event horizon, and when you enter into it, you spaghetti-fy like a thin long noodle, and can never come out of it.

Later on, the discovery of quasi-stellar radio sources (quasars) shook the world of black hole research. Upon investigation of the spectrum emissions, a redshift in the shifting spectrum with the h-spectrum hidden in it was found, indicating that these quasars were moving away from the earth in astounding speeds. The scientists of that time assumed that energy due to gravity around these quasars may be causing it to roam around at such great speeds. This sprouted up another possibility: supermassive black holes.

Upon a lot of research, scientists found out numerous supermassive



The Sound of Two Black Holes Colliding, courtesy of LIGO Caltech

“ ”

...
and the universe said you are *not*
alone

and the universe said you are *not*
separate from every other thing

and the universe said you are the
universe tasting itself, talking to
itself, *reading its own code*

and the universe said i love you
because ***you are love.***

...

- Julian Gough, "The End Poem"

from the creators:

This volume was bound together by late nights, shared curiosity, and a collective desire to look a little closer at the mechanics of whatever is happening around us. We deeply thank you for reading the inaugural edition of entangled.

Until the next observation,
the editorial board

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why entangled?

With “entangled”, this connection should not end when you reach the last page. Instead, it should spark your thoughts and inspire you to reach outward, continuing the shared journey of exploring the science that ultimately shapes our *reality*.

“ ”



connect with $\psi(t)$.

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